machine learning & knowledge extraction

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Knowledge-Guided Physics-Informed Hybrid Learning Framework for Uncertainty-Aware Digital Twin Modeling of Nonlinear Thermal Power Systems

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Abstract

Reliable digital twins of complex nonlinear systems require not only high predictive accuracy but also physical consistency, robustness under degraded operating conditions, and explicit uncertainty handling. Purely data-driven models often suffer from poor generalization, unphysical behaviors, and limited interpretability when facing noisy measurements and unseen operating conditions. This paper introduces a knowledge-guided physics-informed hybrid learning framework that integrates recurrent neural networks with Unscented Kalman Filter (UKF) state estimation and embedded thermodynamic constraints within a unified uncertainty-aware architecture. The proposed PI-LSTM-UKF framework achieves competitive predictive accuracy and improved physical consistency relative to the residual-learning hybrids by tightly integrating physics-informed recurrent learning, thermodynamic constraints, and sequential UKF state estimation. While the UKF provides robust recursive correction under noisy measurements during closed-loop operation, the physics-informed Long Short-Term Memory (PI-LSTM) learns nonlinear corrections and long-term dynamics that cannot be captured by the linear model alone. The proposed framework is systematically benchmarked against a hierarchy of seven modeling approaches, including Dynamic Mode Decomposition with control (DMDc), Sparse Identification of Nonlinear Dynamics (SINDy), and residual-learning variants based on Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM). High-fidelity Simscape simulations of a Rankine-cycle steam turbine system are used as a challenging simulation-based case study. Results show that the knowledge-guided hybrid approach achieves competitive predictive accuracy, improved physical consistency, and robust performance under an unseen load profile, severe thermodynamic degradation, valve hysteresis, and substantially elevated sensor noise. The framework provides a promising simulation-based foundation for uncertainty-aware digital twins of nonlinear thermal power systems. Validation using operational plant data remains necessary before its application to real-time monitoring and predictive maintenance.

Keywords: knowledge-guided learning; physics-informed machine learning; hybrid expert system; digital twin; LSTM; Unscented Kalman Filter; uncertainty-aware modeling; nonlinear thermal power systems

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Academic Editor: Mehmed Kantardzic

Received: 25 May 2026
Revised: 7 August 2026
Accepted: 10 August 2026
Published: 13 August 2026

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1. Introduction

Reliable real-time modeling of complex nonlinear industrial energy systems remains a fundamental challenge across many industrial domains. Such systems frequently exhibit strong nonlinear coupling, multiphysics interactions, actuator hysteresis, and time-varying degradation, which complicate the development of accurate and physically consistent digital twins under noisy measurements and unseen operating conditions [1,2].

Purely physics-based simulators are thermodynamically rigorous but often computationally prohibitive for real-time applications [3]. Purely data-driven models may achieve competitive accuracy on seen data, yet they frequently lack physical consistency, generalize poorly to new conditions, and offer limited interpretability [4,5]. These limitations motivate the development of knowledge-guided physics-informed hybrid learning frameworks that systematically combine domain knowledge (embedded physical constraints) with data-driven learning and sequential state estimation.

Achieving net-zero emissions by 2050 requires large-scale renewable integration and increased flexibility of thermal power plants [6]. Thermal steam power plants based on the Rankine cycle remain a major source of electricity worldwide [1]. They provide a significant share of global installed capacity and support stable base-load operation under fluctuating and uncertain demand. These plants operate by heating, expanding, condensing, and pumping water. These plants can reach thermal efficiencies of 30–45% under supercritical conditions. However, their dynamic behavior is strongly nonlinear. During load-following operation, these nonlinearities complicate control and may reduce efficiency, increase emissions, and accelerate component degradation when fast and accurate dynamic models are unavailable [1,7]. A digital twin (DT) is employed to identify the operating conditions of the steam power plant required to reach the desired flexibility and optimal operation [7,8].

In this paper, we propose a knowledge-guided physics-informed hybrid learning framework that integrates recurrent neural networks with Unscented Kalman Filter (UKF) state estimation and embedded thermodynamic constraints within a unified uncertainty-aware architecture. The framework is designed as a general methodology for nonlinear thermal power systems. We evaluate it on a high-fidelity Rankine-cycle steam turbine system as a challenging and representative simulation-based case study. Steam turbines exhibit rich nonlinear dynamics, two-phase flow, actuator nonlinearities, and strong sensitivity to operating conditions, making them a suitable testbed for assessing robustness and generalization.

Recent research has explored hybrid modeling methods that combine physics-based structure with the flexibility of data-driven learning [2,4], such as machine learning (ML) [4,5], Numerical algorithms for Subspace State-Space System Identification (N4SID) [9], Neural Networks (NN) [5], Unscented Kalman Filter (UKF) [10], Dynamic Mode Decomposition with Control (DMDc) [11], and Sparse Identification of Nonlinear Dynamics (SINDy) [12,13] approaches.

Each of these methods has limitations when used in isolation. Linear techniques such as N4SID and DMDc assume reduced-order linear dynamics and may not capture strong nonlinear coupling effects [14]. SINDy can discover sparse nonlinear dynamics but may leave residual errors when the underlying nonlinear interactions are complex or not well-represented as sparse combinations within a predefined function library [13–15].

The proposed PI-LSTM-UKF framework is systematically benchmarked against a hierarchy of seven modeling approaches, including linear reduced-order identification (DMDc), sparse nonlinear identification (SINDy), residual-learning neural networks (CNN and LSTM), and the full physics-informed hybrid model. Using high-fidelity Simscape simulations with realistic sensor noise and deliberately introduced degradation, this study shows that the proposed framework achieved competitive predictive accuracy and improved

physical consistency relative to the residual-learning hybrids. It also generalizes effectively to unseen load profiles and thermodynamic degradation while providing uncertainty-aware state estimation. The framework embeds thermodynamic constraints directly into the training objective through a two-stage physics-informed training protocol [16].

Main Contribution and Framework Overview

The main contributions of this work are summarized as follows:

- A knowledge-guided physics-informed hybrid learning framework is proposed that combines embedded thermodynamic constraints, recurrent neural learning, and Unscented Kalman Filter (UKF) sequential state estimation for uncertainty-aware modeling of nonlinear thermal power systems.
- A systematic hierarchy of seven modeling variants is developed, including DMDc, SINDy, residual CNN and LSTM hybrids, and the complete PI-LSTM-UKF framework, enabling a controlled analysis of the contribution of each modeling component.
- A two-stage physics-informed training protocol is introduced to enforce thermodynamic consistency while improving predictive performance under noisy measurements and unseen operating conditions.
- Comprehensive robustness evaluation is conducted using a high-fidelity Simscape Rankine-cycle steam turbine model under an unseen load profiles, thermodynamic degradation, valve hysteresis, and elevated sensor noise. The evaluation is presented as a simulation-based stress test rather than industrial validation.
- An uncertainty-aware digital twin formulation is presented that integrates hybrid learning with sequential Bayesian filtering and provides both point predictions and calibrated prediction intervals.

The core technical contribution of this work lies in the integration of three tightly coupled components within a single digital twin architecture: (i) a physics-informed LSTM trained using a two-stage learning protocol, (ii) replacement of the analytical state-transition model within the UKF by the learned physics-informed dynamics, and (iii) systematic isolation of the contribution of each modeling component through a hierarchy of seven model variants. Unlike recent physics-informed recurrent models that primarily enforce differential-equation constraints during training [17], the proposed framework combines constrained recurrent learning with recursive Bayesian filtering, enabling physically consistent prediction together with sequential uncertainty-aware state estimation.

Furthermore, the systematic evaluation across the seven modeling variants provides insight into the individual and combined contributions of each component within the proposed framework, helping to clarify the respective roles of physics-informed learning, recurrent temporal modeling, and recursive Bayesian filtering.

This combination is particularly valuable for industrial digital twins [18] because purely data-driven models can violate physical constraints under extrapolation, whereas purely physics-based or linear Kalman-filter approaches often struggle to represent the strong nonlinearities and long-term temporal dependencies characteristic of thermal power systems. By embedding thermodynamic constraints during training and employing the learned physics-informed dynamics as the UKF state-transition model, the proposed framework provides a practical balance between physical consistency, nonlinear modeling capability, and uncertainty-aware state estimation.

The remainder of this paper is organized as follows: Section 2 is the literature review, Section 3 describes the methodology, Section 4 presents the results, Section 5 discusses the results, and Section 6 concludes with implications and future directions.

2. Literature Review

Digital twins have emerged as a powerful paradigm for real-time monitoring, fault diagnosis, predictive maintenance, and optimization in complex industrial systems. Recent reviews highlight their application not only in thermal power plants but also in integrated renewable systems combining hydro, wind, and solar power [18]. However, developing accurate and reliable digital twins for nonlinear industrial energy systems remains challenging, particularly when high predictive performance must be accompanied by physical consistency, interpretability, and robustness under uncertainty. While many existing implementations focus on simplified scenarios, steady-state operation, or small-scale systems, utility-scale industrial processes often involve multiscale dynamics, strong nonlinear coupling, actuator hysteresis, and time-varying degradation [8,17].

Existing approaches to digital twin modeling can be broadly categorized into physics-based, data-driven, and hybrid methods. Each category offers distinct advantages but also exhibits limitations in terms of knowledge extraction, generalization, and physical consistency, key concerns for machine learning applications in industrial domains.

2.1. Physics-Based Models for Complex Industrial Systems

Physics-based models rely on first-principles conservation laws and thermodynamic formulations such as IAPWS-IF97. They are widely implemented in platforms like Simscape [19], which supports detailed component-level modeling with multiphase flow. Open-source tools such as OpenModelica [20–22] and ThermoPower also enable modular modeling of full process cycles.

While these models inherently respect physical laws, they require extensive parameterization and often degrade under unmodeled effects such as sensor drift, hysteresis, and component aging [2]. Lumped-parameter models offer computational efficiency and robustness in steady state [23], but their accuracy typically deteriorates during fast transients. Finite-volume and moving-boundary models improve transient fidelity [22], yet they remain parameter-intensive and difficult to maintain under degradation. High-fidelity CFD approaches provide detailed insight but are computationally prohibitive for real-time digital twin applications [24].

From a machine learning perspective, purely physics-based models offer strong domain knowledge but limited adaptability and poor scalability for online learning or knowledge extraction from operational data.

2.2. Data-Driven Approaches and Knowledge Extraction in System Identification

Data-driven methods learn system behavior directly from measurements, offering greater flexibility and reduced reliance on detailed first-principles modeling. Linear techniques such as N4SID and DMDc provide low-order, interpretable representations suitable for control-oriented applications [25–27]. Sparse identification methods, particularly SINDy, extend this capability by discovering compact nonlinear governing equations from data, representing a powerful form of knowledge extraction [13].

Deep learning architectures, especially CNN-LSTM models, have demonstrated strong predictive performance in industrial settings, often achieving low MAPE on key variables [28,29]. Time-series machine learning techniques, including recurrent and convolutional models, have been successfully applied to modeling and identification of nonlinear dynamic systems with effects such as friction and hysteresis [30]. However, purely data-driven models frequently suffer from poor generalization, unstable performance under distribution shift ($0.6 < R^2 < 0.8$), and a tendency to produce physically inconsistent predictions [31]. Their accuracy is highly sensitive to data quality, noise, and operating

condition changes [17,32]. Moreover, the lack of embedded physical constraints limits both interpretability and reliability in safety-critical applications.

Recent advances in sparse and symbolic regression have improved the knowledge extraction capabilities of data-driven methods. Nevertheless, standalone application of these techniques often struggles with strong nonlinear coupling, measurement noise, and the need for uncertainty quantification in real-time settings.

2.3. Hybrid and Physics-Informed Digital Twins

Hybrid approaches that combine physics-based models with data-driven corrections have gained significant attention for balancing physical consistency with modeling flexibility [5,7]. These methods retain the nonlinear structure provided by first principles while using machine learning to compensate for unmodeled dynamics and improve accuracy [33].

Physics-informed machine learning has emerged as a particularly promising direction, embedding thermodynamic or physical constraints directly into the learning process [7]. Such approaches improve generalization and reduce the risk of unphysical predictions. More recently, physics-informed generative models have also been investigated. Zhu et al. [34] proposed a physics-informed variational autoencoder (PI-VAE) framework for modeling power plant thermal systems by incorporating thermodynamic constraints directly into the VAE training loss, achieving improved physical consistency with limited training data. However, many existing hybrid and physics-informed models still lack tight integration with sequential state estimation (e.g., filtering) and robust mechanisms for uncertainty propagation. Furthermore, systematic comparisons that isolate the contribution of individual components, such as sparse identification, residual learning, and physics regularization, remain limited, especially under realistic industrial conditions involving noise, degradation, and unseen transients [4,35].

Recent contributions in the machine learning and knowledge extraction have advanced data-driven conceptualizations of digital twins [36] and demonstrated the value of explicitly embedding physical priors into deep learning architectures [37]. Similarly, physics-informed neural architectures have been successfully applied to complex engineering systems with dynamic boundary conditions [38]. These works highlight the growing interest in hybrid physics-ML approaches within the machine learning community, motivating the development of tightly integrated frameworks that combine recurrent learning, embedded thermodynamic constraints, and sequential state estimation.

Despite notable progress, several challenges persist for machine learning-based digital twins of nonlinear systems: (i) achieving robust physical consistency without sacrificing predictive performance, (ii) enabling effective knowledge extraction from both data and domain physics, (iii) providing uncertainty-aware prediction behavior, and (iv) maintaining computational efficiency suitable for real-time deployment. These gaps motivate the development of knowledge-guided physics-informed hybrid frameworks that unify recurrent learning, embedded physical constraints, and sequential filtering within a single architecture.

3. Methodology

To evaluate the proposed knowledge-guided physics-informed hybrid learning framework for nonlinear industrial energy systems, we conduct a comprehensive experimental study using high-fidelity simulation data. The steam turbine Rankine cycle system is employed as a challenging and representative simulation-based case study, characterized by strong nonlinearities, multiphysics coupling, actuator hysteresis, and sensitivity to operating conditions. This choice allows rigorous assessment of the framework's generalization, physical consistency, and robustness under realistic industrial scenarios.

The evaluated models are treated as independent benchmark and hybrid modeling frameworks rather than sequential modules within a single operational architecture. They share a common data-preparation pipeline solely to ensure a fair comparison, as shown in Figure 1.

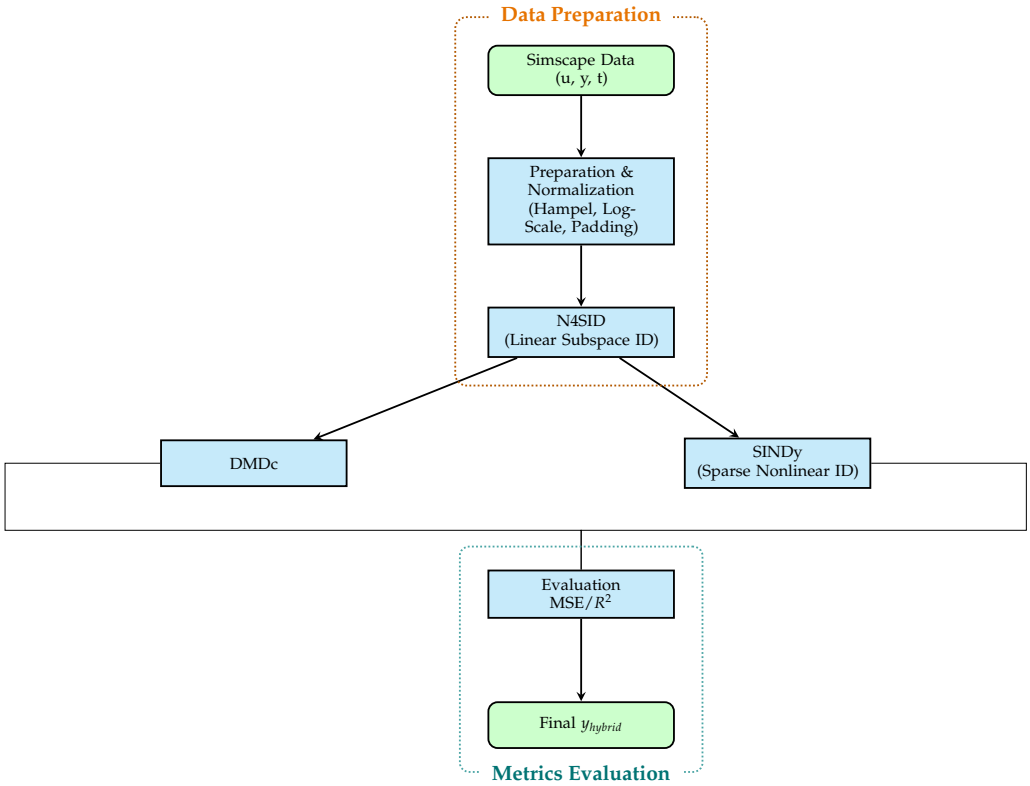


Figure 1. Workflow of the baseline modeling framework used for benchmarking. High-fidelity Simscape data are preprocessed and used to identify analytical DMDc and delay-SINDy-c models. The resulting predictions are evaluated using identical performance metrics before introducing residual-learning or physics-informed enhancements.

The dataset was generated using a high-fidelity Simscape model of the steam turbine system under a wide range of operating conditions, including load variations and disturbances. The reference model was implemented in MATLAB (Simscape R2024a) utilizing the Thermal Liquid and Foundation libraries together with the IAPWS-IF97 steam property package. The complete Rankine cycle includes a drum-type boiler, superheater, reheater, high-pressure turbine with extraction, condenser, and feedwater pump. All control valves were modeled as equal-percentage valves with realistic hysteresis and saturation limits. The model captures strong nonlinear dynamics arising from two-phase flow, valve characteristics, and thermodynamic coupling (Table 1).

Table 1. Model parameters.

Model Inputs	Description/Unit	Model Outputs	Description/Unit
Boiler drum level	L_{Boiler} 700 mm	Actuator position for high pressure valve	V_{HP} in % 85%
Steam pre-heat level	L_{heater} 200 mm	Actuator position for extraction valve	$V_{\text{extraction}}$ 55%
Heat input rate	$\dot{Q}_{\text{heat}}$ 1500 MWth	Feedwater output flow rate	$\dot{Q}_{\text{FW pump}}$ 450 kg/s
Super-heat valve position	$V_{\text{super-heat}}$ 85%	System efficiency	η_{system} 0.39 (39%)
Re-heat valve position	$V_{\text{re-heat}}$ 80%	Electrical power output	P_{out} MW $\pm$ 1.2 MW
Grid power demand	P_{demand} MW		

Simulation scenarios were generated at 10^5 s, covering a wide range of load ramps, step changes, and valve perturbations. The full dataset was chronologically split into training (70%), validation (15%), and test (15%) sets while preserving temporal order and avoiding information leakage, as summarized in Table 2.

Table 2. Simulation and data generation parameters.

Simulation and Data Generation Parameters	Value
Hardware	Intel Core i7, 32 GB RAM, CPU only
Simulation environment	MATLAB Simscape/Simulink
Sampling time	1 s
Simulation scenarios	8 scenarios (same plant)
Simulation horizon	10^5 s (training/validation/test segments extracted)
Simulation run time	Approximately 9–12 h
Total Samples	100,000
Training/validation/test split	70%/15%/15%
Main excitation types	Load ramps, step changes, valve perturbations
Disturbance amplitude range	±20–40% around nominal values

Main evaluation record: one continuous 100,000-sample scenario. The other seven scenarios were used only for development and qualitative checks.

All preprocessing steps including normalization, scaling, and outlier handling were applied after data partitioning. All normalization parameters (minimum and maximum values for log-normalization) were computed exclusively on the training set and then applied to validation and test sets. This strictly prevents any information from the validation or test sets from influencing the training process, which is critical for reliable assessment of generalization in machine learning models for dynamical systems. The resulting datasets were then used to train and evaluate all compared models under consistent conditions.

Figures 1–3 illustrate the overall experimental pipelines, including data generation, preprocessing, model identification, residual correction, physics-informed training, and evaluation.

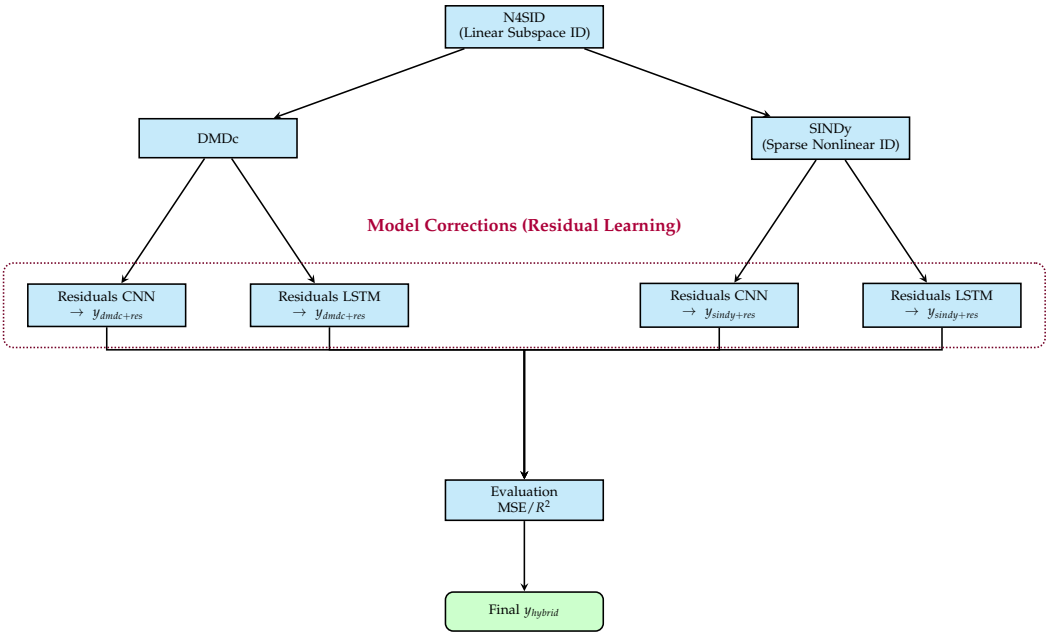


Figure 2. Residual-learning architectures built upon the analytical DMDc and delay-SINDy-c baselines. CNN and LSTM networks learn one-step prediction residuals using only historical information, producing four hybrid benchmark models (DMDc + CNN, DMDc + LSTM, SINDy + CNN, and SINDy + LSTM) evaluated under an identical leakage-free protocol.

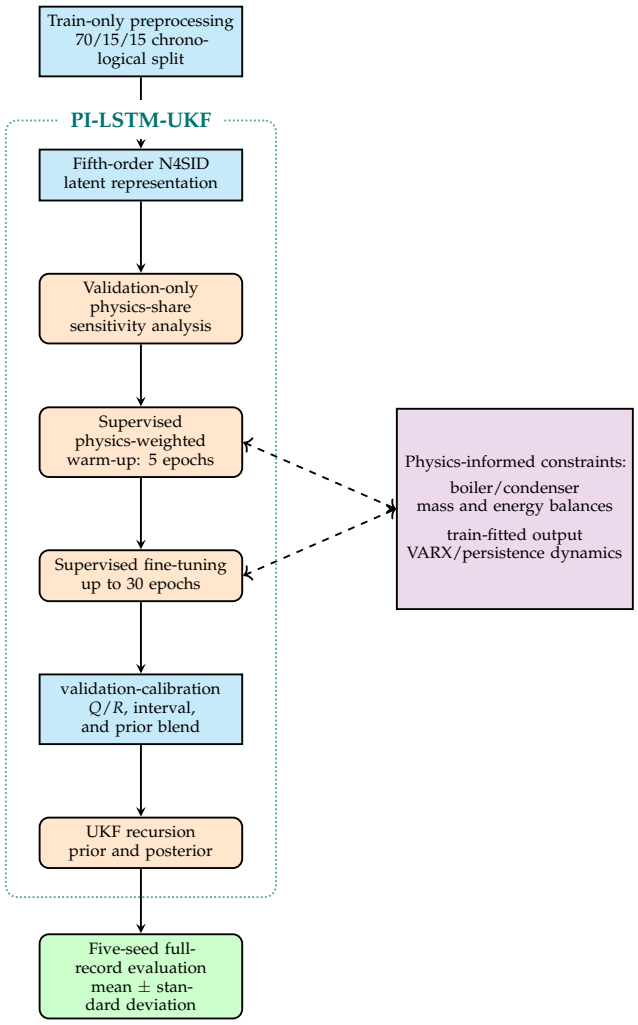


Figure 3. Leakage-controlled PI-LSTM-UKF development workflow. Training-only normalization, chronological data partitioning, validation-based physics-weight selection, two-stage physics-informed training, UKF calibration, and final evaluation are performed sequentially to prevent information leakage. Prior prediction and posterior filtering are evaluated separately after all model-selection decisions are fixed. Solid arrows represent the sequential workflow, whereas dashed bidirectional arrows indicate the incorporation of the physics-informed constraints during the warm-up and fine-tuning stages.

3.1. High-Fidelity Reference Model

The experimental pipeline begins with the generation of high-fidelity simulation data from eight independent simulations of the same Simscape plant model, each representing a distinct operating scenario with different excitation patterns (e.g., step, ramp, and composite variations) and thermodynamic degradation conditions. The scenario is simulated to produce 100,000 samples using IAPWS-IF97 thermodynamic properties [19,39]. This high-fidelity simulator serves as a reliable virtual testbed for developing and benchmarking data-driven and hybrid models when real plant data is unavailable due to confidentiality, safety, or operational constraints.

Table 1 summarizes the plant input and output variables along with their physical units and nominal operating ranges. These variables are normalized and used within Simscape to solve the conservation equations of mass, momentum, and energy across all major components of the Rankine cycle, including the boiler, turbines, feedwater pumps, condenser, and regenerative heaters [40]. The simulator captures realistic industrial effects such as valve hysteresis, actuator saturation, and strong thermodynamic coupling. The principal model variables are summarized in Table 1.

Eight operating scenarios were generated by executing the same Rankine-cycle Simscape plant model under different operating conditions and input-signal patterns (different load trajectories, valve excitation patterns, and mild thermodynamic variations). All scenarios were generated using the same plant model and differ only in excitation and degradation conditions.

The principal quantitative results reported in this paper (including all tables of MSE, R^2, ablation studies, physics-share sensitivity, and the five-seed analysis) were obtained exclusively from **one continuous 100,000-sample record** sampled at 1 s. This record was partitioned chronologically into 70,000 training samples, 15,000 validation samples, and 15,000 untouched test samples.

The remaining scenarios were used only for model development, qualitative inspection, and complementary stress testing. They were *not* included in the calculation of the principal performance metrics or the five-seed statistical analysis reported in Section 4. Consequently, the five-seed experiment measures sensitivity to stochastic neural network initialization on a fixed simulation record and does not constitute validation across independent operating scenarios.

3.2. State-Space Representation

To enable systematic learning and evaluation of the proposed knowledge-guided hybrid framework, we represent the steam turbine system in state-space form. The system input and output vectors at time step k, corresponding to the variables listed in Table 1, are defined as $\mathbf{u}_k \in \mathbb{R}^6$ and $\mathbf{y}_k \in \mathbb{R}^5$:

$$\mathbf{u}_k = [L_{\text{Boiler}}, L_{\text{heater}}, \dot{Q}_{\text{heat}}, V_{\text{super-heat}}, V_{\text{re-heat}}, P_{\text{demand}}]^\top,$$
$$\mathbf{y}_k = [V_{\text{HP}}, V_{\text{extraction}}, \dot{Q}_{\text{FW pump}}, \eta_{\text{system}}, P_{\text{out}}]^\top \quad (1)$$

Data preparation follows standard machine learning best practices for time-series modeling. Zero-mean Gaussian noise ($\sigma = 0.05$) was added to the outputs to simulate realistic sensor noise and measurement uncertainty. A Hampel filter (window size 5, threshold 3 $\times$ MAD) was applied for robust outlier correction while preserving short-term dynamics [41]. All preprocessing steps, including log-normalization and zero-padding, were performed after data splitting to strictly prevent information leakage from validation and test sets. Although the full dataset contains approximately 100 thousand samples, after padding, training was performed on chronologically split sequences with efficient mini-batch processing. All experiments were conducted on CPU only.

A Hankel matrix was constructed from the preprocessed outputs to support model order estimation. All signals were log-normalized with zero-padding ($N_{\text{steps}} = 10$) for stable initialization [42]:

Table 2 reports simulation settings and hardware specifications.

$$\tilde{u}_i(k) = \log\left(\frac{u_i(k) - \min(u_i)}{\max(u_i) - \min(u_i)} + 1\right),$$
$$\tilde{y}_j(k) = \log\left(\frac{y_j(k) - \min(y_j)}{\max(y_j) - \min(y_j)} + 1\right) \quad (2)$$

An initial linear state-space model is identified using N4SID [9]. This method employs block-Hankel matrices, projections, and SVD to estimate (A, B, C, D) without prior parameterization or local minima issues, making it suitable for multivariable systems. The N4SID algorithm returns a discrete-time linear state-space model of order $n = 5$.

3.3. Evaluated Model Variants

To systematically analyze the contribution of each modeling component, we evaluate seven progressively sophisticated model variants. These variants form a structured hierarchy ranging from linear baselines to the full proposed knowledge-guided physics-informed framework Figures 1–3:

1. N4SID-state DMDc [11] (linear reduced-order open-loop baseline).
2. Delay-SINDy-c [12] (sparse nonlinear one-step baseline).
3. Proposed DMDc + CNN.
4. Proposed DMDc + LSTM.
5. Proposed SINDy + CNN.
6. Proposed SINDy + LSTM.
7. Proposed PI-LSTM-UKF (full knowledge-guided physics-informed hybrid framework).

The first two variants serve as analytical baselines. The next four models add residual correction using CNN or LSTM networks under a strict one-step-ahead protocol. The final variant introduces physics-informed recurrent learning together with recursive UKF correction. This hierarchical design isolates the individual contributions of linear identification, sparse knowledge extraction, residual learning, physics-informed regularisation, and sequential filtering.

The following subsections describe these variants in detail.

3.4. Fair Benchmarking Protocol

To ensure a fair and reproducible comparison, all neural network-based models (CNN and LSTM residual correctors and the PI-LSTM) were trained using the Adam optimizer with a step-decay learning rate schedule and L2 regularization. Early stopping with a validation patience of 8 epochs was applied to the residual correction models based on the validation loss, with a maximum training limit of 50 epochs. The network achieving the lowest validation loss was retained. The PI-LSTM-UKF used a 5-epoch physics-weighted warm-up followed by at most 30 epochs of fine-tuning, with early stopping on the validation-selection partition. Physics regularization in both stages is controlled by target shares s_{pre} and s_{fine}. DMDc and SINDy are algebraic identification methods with relatively few hyperparameters. Their implementations followed standard formulations based on truncated SVD and sequential thresholded least squares, respectively. Training was repeated over 5 independent random seeds to report the mean and standard deviation, providing statistical robustness to the comparison.

3.5. Baseline Modeling

To ensure a fair and reproducible comparison, all baseline models were developed under an identical leakage-free one-step-ahead prediction protocol using the same 100,000-sample dataset sampled at 1 s. The data were partitioned chronologically into 70% training, 15% validation, and 15% testing subsets. All normalization statistics, state-space identification, model parameters, and neural network weights were estimated exclusively from the training data. The validation set was used solely for hyperparameter selection, regularization, and early stopping, whereas the test set remained untouched until the final evaluation. Consequently, no information from the validation or test partitions influenced model training.

Two fundamentally different system-identification approaches were selected as benchmark models:

- **N4SID-state DMDc**, representing a reduced-order linear state-space predictor [14,43];
- **Delay-SINDy-c**, representing a sparse nonlinear discrete-time predictor [15].

For the first benchmark, N4SID identifies a stable fifth-order state-space realization from the training data. The resulting latent states are subsequently standardized using training-set statistics to improve numerical conditioning before fitting a Dynamic Mode Decomposition with Control (DMDc) model via ridge regression [14,43]. The resulting control-affine linear model is propagated in an open loop to generate the baseline trajectory $\hat{y}_{\mathrm{DMDc}}(k)$.

The second benchmark employs a discrete delay-SINDy-c formulation based on the Sparse Identification of Nonlinear Dynamics (SINDy) framework [15]. A delay-embedded feature library is constructed directly from the normalized inputs and outputs and comprises delayed outputs, delayed inputs, a constant term, and optional quadratic interactions. Sparse model coefficients are identified using sequential thresholded least squares with ridge regularization [15]. The embedding delay, polynomial degree, sparsity threshold, and ridge coefficient are selected exclusively using the validation set. The resulting sparse discrete-time mapping generates the one-step baseline trajectory $\hat{y}_{\mathrm{SINDy}}(k)$. Those model parameters are presented in Table 3.

To evaluate whether residual learning can compensate for the remaining modeling errors of these analytical baselines, lightweight one-dimensional CNN and LSTM residual predictors are trained under the identical leakage-free one-step prediction protocol [16,17]. This produces four hybrid benchmark models: DMDc + CNN, DMDc + LSTM, SINDy + CNN, and SINDy + LSTM.

3.5.1. Dynamic Mode Decomposition with Control (DMDc)

The first benchmark employs an N4SID-state DMDc model that constructs a reduced-order control-affine linear representation from the latent state trajectories identified by N4SID [14,43]. Following fifth-order state-space identification on the training data, the latent states are standardized using training-set statistics. DMDc is subsequently estimated by solving the ridge-regularized least squares problem

$$\theta^* = \arg\min_{\theta} \left\| X_2 - \begin{bmatrix} X_1 \\ U \end{bmatrix} \theta^\top \right\|_F^2 + \lambda \|\theta\|_F^2, \tag{3}$$

where

$$\theta^* = \begin{bmatrix} A_{\mathrm{dmd}} & B_{\mathrm{dmd}} \end{bmatrix}. \tag{4}$$

Here, X_1 and X_2 denote consecutive standardized latent-state snapshots, while U contains the corresponding control inputs. The identified dynamics are represented by

$$x_{k+1} = A_{\mathrm{dmd}} x_k + B_{\mathrm{dmd}} u_k + c, \tag{5}$$

where c denotes an affine offset estimated from the training data.

To guarantee stable open-loop prediction, the spectral radius of A_{dmd} is constrained below 0.999, after which the affine offset is recomputed to preserve the mean operating point of the training data. The ridge regularization coefficient is selected using the validation set, and the finalized model is evaluated only once on the held-out test set as an open-loop one-step predictor. The resulting reduced-order representation provides a computationally efficient linear approximation of the dominant system dynamics while serving as a benchmark for subsequent nonlinear hybrid models [14,43].

Table 3. Implementation and hyperparameter settings for the baseline and residual models.

Parameter	Value
Dataset size	100,000 samples
Sampling interval	1 s
Data partition	Chronological 70/15/15 train-validation-test split
Measurement-noise level	Gaussian noise with standard deviation equal to 5%
Normalization	Feature-wise standardization using training-set mean and standard deviation only
Residual lookback window (w)	20 previous samples, $k-w, \ldots, k-1$
Prediction protocol	Rolling one-step-ahead prediction using measurements available only through sample $k-1$
N4SID-state DMDc	
DMDc ridge candidates (λ_{DMDc})	$\{10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$
DMDc stability constraint	$\rho(A_{\mathrm{dmd}}) \leq 0.999$
DMDc model selection	Ridge coefficient selected using validation performance
Delay-SINDy-c	
Candidate embedding delays (q)	$\{2, 5, 10, 20\}$ samples
Candidate polynomial degrees	$\{1, 2\}$
Candidate sparsity thresholds	$\{10^{-3}, 10^{-2}, 10^{-1}\}$
Candidate ridge coefficients	$\{10^{-6}, 10^{-4}, 10^{-2}\}$
Sequential thresholding iterations	20
Hyperparameter-search stride	4 samples
Sparsity-complexity penalty	10^{-4}
SINDy model selection	Delay, polynomial degree, sparsity threshold, and ridge coefficient selected using validation performance
CNN and LSTM residual models	
Residual-model input features	Normalized inputs, measured outputs, baseline predictions, and previous baseline residuals
CNN architecture	Temporal convolution layers with 32 filters of kernel size 5 and 64 filters of kernel size 3, dilation factor 2 in the second convolution, batch normalization, ReLU activation, 0.2 dropout, global average pooling, and a five-output fully connected layer
LSTM architecture	One LSTM layer with 128 hidden units and last-output mode, followed by 0.2 dropout and a five-output fully connected layer
Optimizer	Adam [44]
Maximum epochs	20
Mini-batch size	128–256
Initial learning rate	1×10^{-3}
L_2 regularization	1×10^{-4}
Gradient threshold	1
Validation patience	8 epochs
Network selection	Network parameters corresponding to the minimum validation loss
Training-data shuffling	Every epoch

3.5.2. Sparse Nonlinear Identification via SINDy

The second benchmark employs a delay-SINDy-c model that identifies a sparse discrete-time nonlinear mapping directly from the normalized system inputs and outputs [15]. Unlike the classical continuous-time formulation, which requires numerical differentiation of the system states, the adopted discrete implementation operates directly on delay-embedded measurements, thereby avoiding derivative estimation and improving robustness to measurement noise.

The discrete nonlinear mapping is expressed as

$$\hat{y}(k) = \Theta(y(k-q:k-1), u(k-q:k-1))\Xi, \tag{6}$$

where $\Theta(\cdot)$ denotes the delay-embedded candidate library consisting of delayed outputs, delayed inputs, a constant term, and optional quadratic monomials, while Ξ is the sparse coefficient matrix identified using sequential thresholded least squares with ridge regularization [15].

The embedding delay q, polynomial degree, sparsity threshold, and ridge regularization coefficient are selected exclusively using validation performance. The resulting sparse discrete-time representation retains the interpretability of the original SINDy framework while providing a practical one-step predictor for nonlinear system dynamics. After all hyperparameters are fixed, the identified model is evaluated once on the held-out test set to generate the baseline trajectory $\hat{y}_{\text{SINDy}}(k)$.

Although the sparse representation substantially improves the modeling of nonlinear interactions compared with purely linear dynamics, prediction errors may still arise when complex system behavior cannot be represented adequately by the selected candidate library [15]. These remaining errors motivate the subsequent residual-learning stage based on CNN and LSTM networks.

3.6. Proposed CNN-Based Residual Correction

Although the DMDc and delay-SINDy-c models capture the dominant linear and sparse nonlinear dynamics of the steam turbine, systematic prediction errors remain because of unmodeled nonlinear interactions and operating-point variations. To compensate for these residual errors, a one-dimensional Convolutional Neural Network (CNN) is employed as a lightweight residual predictor following the residual-learning strategy adopted in recent hybrid digital twin models [16,17].

For each baseline model, the residual signal is defined as

$$r(k) = y(k) - \hat{y}_{\text{base}}(k), \tag{7}$$

where $y(k)$ denotes the measured output and $\hat{y}_{\text{base}}(k)$ represents the corresponding prediction obtained from either the N4SID-state DMDc or the delay-SINDy-c baseline. The residuals are standardized using the mean (μ_r) and standard deviation (σ_r) computed exclusively from the training partition.

The CNN receives a fixed-length sliding window ($w = 20$) composed of normalized historical information available up to time step $k-1$. The input sequence contains the normalized system inputs, measured outputs, baseline predictions, and previously observed residuals, ensuring that only past information is available during prediction. This formulation preserves the one-step-ahead leakage-free evaluation protocol adopted throughout this study.

The CNN residual predictor is expressed as

$$\hat{r}_{\text{CNN}}(k) = f_{\text{CNN}}(X(k), \theta_{\text{CNN}}), \tag{8}$$

where $X(k)$ denotes the input window and θ_{CNN} represents the trainable network parameters.

The corrected prediction is subsequently obtained as

$$\hat{y}_{\text{corr}}(k) = \hat{y}_{\text{base}}(k) + \hat{r}_{\text{CNN}}(k)\sigma_r + \mu_r, \tag{9}$$

where μ_r and σ_r denote the mean and standard deviation of the training residuals, respectively.

Network parameters are optimized using the Adam optimizer [44], while validation data are used exclusively for early stopping and hyperparameter selection. The CNN

residual correction is applied independently to both baseline models, producing the DMDc + CNN and SINDy + CNN hybrid predictors shown in Figure 2. The corresponding implementation details are summarized in Table 3.

Because convolutional filters primarily exploit local temporal correlations, the CNN provides an efficient mechanism for modeling short-term residual dynamics while maintaining a relatively low computational cost.

3.7. Proposed LSTM-Based Residual Correction

To capture longer temporal dependencies that cannot be represented effectively by local convolutional filters, the residual correction stage is alternatively implemented using a Long Short-Term Memory (LSTM) network [45]. The LSTM replaces the CNN while preserving the identical leakage-free training and evaluation protocol, thereby enabling a direct comparison between the two sequence-learning architectures.

The LSTM receives exactly the same input representation as the CNN residual model, namely a sliding window containing normalized historical inputs, measured outputs, baseline predictions, and previously observed residuals. Consequently, any performance differences between the CNN- and LSTM-based hybrid models arise from the underlying sequence-learning mechanism rather than differences in the available information.

The residual predictor is defined as

$$\hat{r}_{\mathrm{LSTM}}(k) = f_{\mathrm{LSTM}}(X(k), \theta_{\mathrm{LSTM}}), \tag{10}$$

where θ_{LSTM} denotes the trainable LSTM parameters.

The corrected output is computed using the same residual reconstruction adopted for the CNN model,

$$\hat{y}_{\mathrm{corr}}(k) = \hat{y}_{\mathrm{base}}(k) + \hat{r}_{\mathrm{LSTM}}(k)\sigma_r + \mu_r, \tag{11}$$

where $\hat{y}_{\mathrm{base}}(k)$ corresponds to the prediction generated by either the N4SID-state DMDc or the delay-SINDy-c baseline.

Training is performed using the Adam optimizer [44], with validation-guided early stopping and the same optimization settings adopted for the CNN model to ensure a fair comparison. The network employs gated memory cells that selectively retain and discard temporal information, allowing delayed system responses and long-range dependencies to be learned more effectively than feedforward architectures [45].

The LSTM residual correction is likewise applied independently to both analytical baselines, yielding the DMDc + LSTM and SINDy + LSTM benchmark models shown in Figure 2. The complete training configuration is provided in Table 3.

Although LSTM-based residual learning generally improves the representation of long-term temporal behavior, the resulting model remains entirely data-driven and therefore does not explicitly enforce the governing thermodynamic relationships of the steam turbine. This limitation motivates the proposed physics-informed LSTM framework introduced in the following subsection, where physical constraints are incorporated directly into the learning process [17].

3.8. Proposed Physics-Informed LSTM Hybrid with UKF (PI-LSTM-UKF)

Physics-informed recurrent models [46,47] and neural network-assisted Kalman filters [48] address complementary aspects of nonlinear prediction and recursive state estimation. Mass- and energy-conserving LSTM architectures have previously been explored, particularly in hydrology and thermal systems, where conservation laws are enforced either by architectural constraints [49] or by adding balance residuals to the loss function [50]. The proposed PI-LSTM-UKF combines a reduced-order N4SID latent representation, a non-

linear recurrent transition model, explicit physical regularization, and recursive Unscented Kalman Filter (UKF) correction. N4SID defines the latent coordinates and linear output map, whereas the LSTM predicts the nonlinear one-step latent transition used during sigma-point propagation. The framework produces both a pre-measurement one-step prior and a post-measurement posterior, which are reported and evaluated separately.

3.8.1. Leakage-Controlled Data Preparation

The five measured outputs, six control inputs, and required Simscape physical variables are resampled to a uniform record of $N = 100{,}000$ observations at a sampling interval of $\Delta t = 1\,\mathrm{s}$. A chronological 70/15/15 training-validation-test split is used. Consequently, the first 70% of the record is used for model fitting, the following 15% is used for validation, and the final 15% is retained as an untouched test set.

The validation block is divided chronologically into two equal subsets. The first half, denoted validation-calibration, is reserved exclusively for UKF covariance calibration, uncertainty-interval scaling, and prior-blend fitting. The second half, denoted val-select, is used for neural network early stopping, validation assessment, and model-selection criteria. No test sample is used to fit, calibrate, or select any part of the proposed framework.

All transformations and normalization statistics are estimated from the training segment only. For any input, output, latent, or physical channel z, normalization is performed as

$$z_k^n = \frac{z_k - \mu_{z,\mathrm{tr}}}{\sigma_{z,\mathrm{tr}}}, \tag{12}$$

where $\mu_{z,\mathrm{tr}}$ and $\sigma_{z,\mathrm{tr}}$ are the training-set mean and standard deviation, respectively. The same frozen statistics are applied to validation-calibration, val-select, and test samples.

3.8.2. PI-LSTM Transition Model

Each causal network input contains the preceding 30 samples of the normalized control inputs and N4SID latent states. For prediction at sample k, the network therefore uses information available only through sample $k-1$.

The recurrent architecture contains an LSTM layer with 128 hidden units and sequence output, followed by 15% dropout, a second LSTM layer with 64 hidden units and last-step output, 10% dropout, and a fully connected output layer. The output dimension is $n_x + 4$, where the first n_x entries predict the next latent state and the remaining four entries form a jointly trained physical conservation head.

The network predicts the absolute next latent state rather than an additive correction to the N4SID transition:

$$\hat{\mathbf{x}}_k^{\mathrm{LSTM}} = \mathcal{F}_{\boldsymbol{\theta}}\left(\{\mathbf{u}^n, \mathbf{x}\}_{k-30:k-1}\right)_{1:n_x}. \tag{13}$$

The corresponding raw LSTM output prior is obtained using the N4SID measurement map:

$$\hat{\mathbf{y}}_k^{\mathrm{LSTM},-} = \mathbf{C}\hat{\mathbf{x}}_k^{\mathrm{LSTM}} + \mathbf{D}\mathbf{u}_k^n + \mathbf{b}. \tag{14}$$

3.8.3. Two-Stage Training Objective

Training consists of a five-epoch supervised physics-weighted warm-up followed by supervised fine-tuning for at most 30 epochs. The warm-up is neither a physics-only nor an unlabeled stage; the latent-state, measured-output, bias, rollout, and physics losses remain active during both stages.

Adam optimization [44] is used with a mini-batch size of 128, an initial learning rate of 5×10^{-4}, and element-wise gradient clipping to the interval $[-1, 1]$. Fine-tuning uses

early stopping on val-select with a patience of five epochs and a minimum improvement tolerance of 10^{-6}.

The implemented data loss is

$$\mathcal{L}_{\text{data}} = \mathcal{L}_x + \lambda_y \mathcal{L}_y + \mathcal{L}_b + \mathcal{L}_{\text{roll}}, \tag{15}$$

where $\mathcal{L}_x$ is the next-latent-state loss, $\mathcal{L}_y$ is the measured-output loss, $\mathcal{L}_b$ penalizes systematic channel bias, and $\mathcal{L}_{\text{roll}}$ is a three-step recursive output-prediction loss.

The output-loss channel weights are

$$\mathbf{w}_y = [1, 1, 4, 1, 1], \tag{16}$$

which gives additional training emphasis to the steam-extraction-valve channel. The coefficient λ_y is calibrated from 256 uniformly spaced training sequences so that the output loss initially contributes a target share of 0.15 relative to the latent-state and output objectives.

Scheduled sampling is used during recursive rollout training. Its maximum probability is 0.20 and is reached linearly over the first 20 epochs.

3.8.4. Physics-Informed Regularization

The physics-informed regularizer combines a first-principles chamber conservation term and a train-fitted measured-output dynamics term. Similar discrete conservation constraints have been used in physics-informed LSTMs for streamflow and thermal systems [17,50], although most prior work relies on continuous-time residuals or architectural mass conservation rather than integrated rates extracted from a high-fidelity simulator.

Boiler and Condenser Conservation

The jointly trained four-output physical head represents the normalized boiler mass, boiler internal energy, condenser mass, and condenser internal energy,

$$\mathbf{z} = \begin{bmatrix} M_{\text{b}} & E_{\text{b}} & M_{\text{c}} & E_{\text{c}} \end{bmatrix}^T, \qquad E_j = M_j u_j. \tag{17}$$

The corresponding Simscape-derived storage rates are represented by

$$\mathbf{f}_{\text{phys}} = \begin{bmatrix} \dot{m}_{\text{b},A} + \dot{m}_{\text{b},B} \\ \Phi_{\text{b},A} + \Phi_{\text{b},B} + \dot{Q}_{\text{b},H} \\ \dot{m}_{\text{c},A} + \dot{m}_{\text{c},B} \\ \Phi_{\text{c},A} + \Phi_{\text{c},B} + \dot{Q}_{\text{c},H} \end{bmatrix}. \tag{18}$$

The flow and heat-rate quantities are integrated on the native variable-step Simscape grid before interval averaging. The discrete physical target is

$$\mathbf{z}_k^{\text{bal}} = \mathbf{z}_{k-1} + \Delta t \, \bar{\mathbf{f}}_{\text{phys},k-1}. \tag{19}$$

Physical states and rates are normalized using separate training-only means and standard deviations. The chamber loss $\mathcal{L}_{\text{ch}}$ compares the normalized balance target with the four auxiliary network outputs. This formulation differs from a continuous-time PINN residual based on automatic differentiation [51] and simplified thermodynamic approximations such as $h = c_p T$ [52].

Measured-Output Dynamics Constraint

A second constraint acts directly on all five measured outputs. For each output channel i, a normalized one-step Vector AutoRegressive with eXogenous inputs (VARX) relation is fitted as

$$\widetilde{y}_{i,k}^{\mathrm{dyn}} = \boldsymbol{\theta}_i^T \begin{bmatrix} \mathbf{y}_{k-1}^n & \mathbf{u}_{k-\ell_i}^n & 1 \end{bmatrix}^T, \qquad \ell_i \in \{0,1\}. \tag{20}$$

The input lag and ridge coefficient are selected from

$$\ell_i \in \{0,1\}, \qquad \gamma_i \in \left\{10^{-6}, 10^{-4}, 10^{-2}, 1\right\}, \tag{21}$$

using the final 10% of the training segment as an internal tuning tail. The selected relation is then refitted using the complete training segment. Persistence is retained as the channel dynamics model whenever it outperforms the fitted VARX relation on the internal training tail.

The output-dynamics channel weights are

$$\mathbf{w}_{\mathrm{dyn}} = [2,1,1,1,2], \tag{22}$$

which gives additional direct prior emphasis to the HP-turbine-valve and power channels.

The complete physical regularizer is

$$\mathcal{L}_{\mathrm{phys}} = \frac{1}{2}\left(\mathcal{L}_{\mathrm{ch}} + \mathcal{L}_{\mathrm{dyn}}\right). \tag{23}$$

Therefore, $\mathcal{L}_{\mathrm{phys}}$ is a composite knowledge-guided objective. The chamber term is derived from first-principles mass and energy conservation, whereas the output-dynamics term is identified exclusively from training data. While mass-conserving LSTMs [49] and energy-balance constraints in recurrent models [17,50] have been studied previously, the present formulation differs by combining a simulator-derived discrete conservation head with data-driven VARX output dynamics, a blended one-step prior, and UKF correction inside a leakage-controlled framework.

3.8.5. Target-Share Physics Weighting

Physics regularization is controlled through target shares rather than fixed raw values of λ_{phys}. For a requested target physics share s, the effective coefficient is computed as

$$\lambda_{\mathrm{phys}}(s) = \frac{s}{1-s} \frac{\mathcal{L}_{\mathrm{cal}}}{\max\left(\mathcal{L}_{\mathrm{phys}}, \epsilon\right)}, \tag{24}$$

and the complete loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{phys}} \mathcal{L}_{\mathrm{phys}}. \tag{25}$$

For physics-share calibration,

$$\mathcal{L}_{\mathrm{cal}} = \mathcal{L}_x + \lambda_y \mathcal{L}_y + \mathcal{L}_b. \tag{26}$$

The effective physics coefficient is calibrated once before the warm-up stage and recalibrated after warm-up before fine-tuning. This stage-wise recalibration accounts for the change in the relative scales of the data and physics losses during optimization [53]. It also means that the selected values 0.20 and 0.50 are target shares, not direct numerical values of λ_{phys}.

3.8.6. Validation-Only Physics-Share Sensitivity Analysis [54]

Physics-share selection is performed before the final full-record model evaluation. To reduce the computational burden without accessing the test partition, a leakage-controlled subset of 5000 chronological samples is used for screening. It contains 3500 training samples, 750 validation-calibration samples, and 750 val-select samples.

The pretraining target shares are

$$\mathcal{S}_{\text{pre}} = \{0, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60\}, \tag{27}$$

and the fine-tuning target shares are

$$\mathcal{S}_{\text{fine}} = \{0, 0.10, 0.20, 0.30, 0.50, 0.60, 0.70, 0.80\}. \tag{28}$$

The resulting 8×8 grid contains 64 target-share combinations. Each combination is evaluated using five predefined random seeds resulting in 320 screening fits.

During sensitivity screening, the UKF covariance scale factors are fixed to one for every candidate so that covariance calibration cannot favor one physics-share pair over another. Prior-blend weights are estimated on validation-calibration, whereas ranking is based only on pre-measurement one-step prior errors on validation-select.

For each target-share pair, the validation score is

$$J_{\text{sel}} = \overline{\text{RMSE}}_{\text{prior}} + 0.25\,\text{RMSE}_{\text{prior,worst}}, \tag{29}$$

subject to the requirement that no channel degrades beyond the permitted factor relative to the corresponding zero-physics reference. As shown in Figure 4, the selection procedure identified a warm-up target share of 0.20 and a fine-tuning target share of 0.50.

This two-stage target-share strategy extends earlier single-coefficient physics weighting schemes commonly used in physics-informed LSTMs [17].

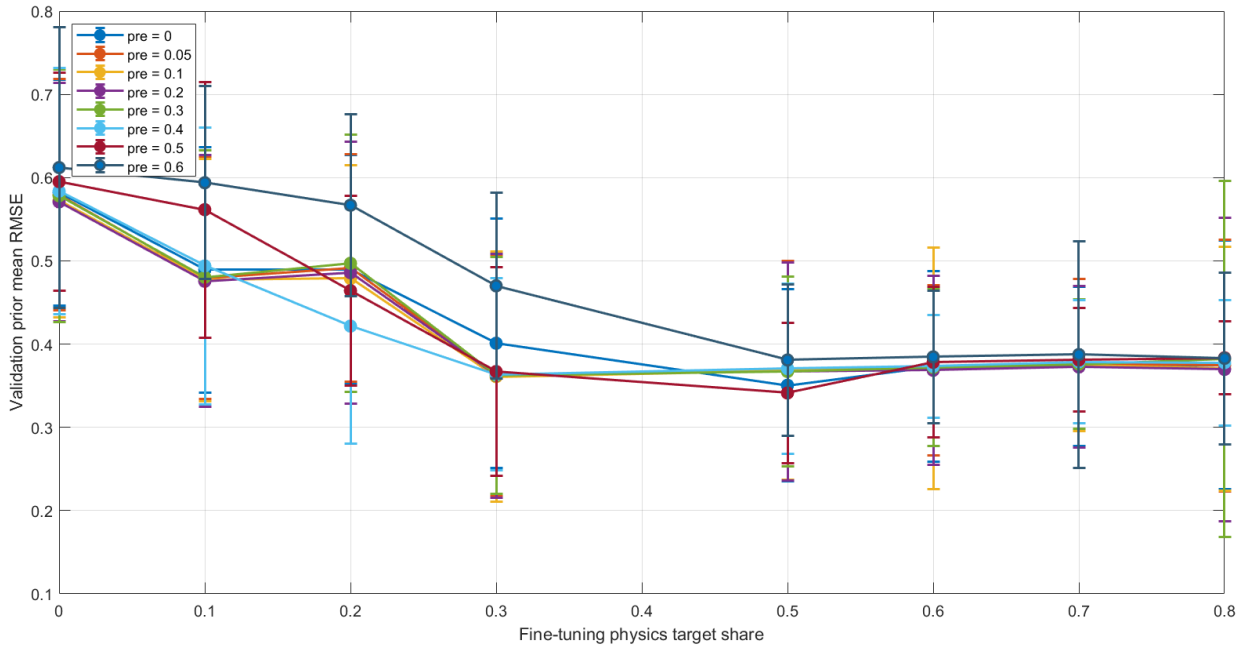


Figure 4. Validation-only physics-share sensitivity analysis. The pretraining and fine-tuning axes represent target physics shares rather than raw λ_{phys} values. Each grid point summarizes five random seeds. The pair $s_{\text{pre}} = 0.20$ and $s_{\text{fine}} = 0.50$ was selected using the composite one-step-prior validation criterion in Equation (29), together with the channel-degradation guard.

3.8.7. UKF Integration and Validation-Only Calibration

The UKF uses the scaled unscented-transform parameters

$$\alpha = 0.1, \qquad \beta = 2, \qquad \kappa = 0. \tag{30}$$

At each sample, the $2n_x + 1$ sigma points replace the latest latent state in the causal 30-sample first-in-first-out history. All sigma-point histories are propagated through the trained LSTM in a single batched forward pass. Their weighted moments define the predicted latent-state mean $\widehat{\mathbf{x}}_k^-$ and covariance $\mathbf{P}_k^-$.

The raw LSTM measurement prediction is

$$\widehat{\mathbf{y}}_k^{\mathrm{LSTM},-} = \mathbf{C}\widehat{\mathbf{x}}_k^- + \mathbf{D}\mathbf{u}_k^n + \mathbf{b}. \tag{31}$$

The innovation covariance and posterior state are

$$\mathbf{S}_k = \mathbf{C}\mathbf{P}_k^-\mathbf{C}^T + \mathbf{R}, \tag{32}$$

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{C}^T\mathbf{S}_k^{-1}, \tag{33}$$

$$\widehat{\mathbf{x}}_k^+ = \widehat{\mathbf{x}}_k^- + \mathbf{K}_k\left(\mathbf{y}_k^n - \widehat{\mathbf{y}}_k^{\mathrm{LSTM},-}\right). \tag{34}$$

A Joseph-form covariance update, covariance symmetrization, and an eigenvalue floor of 10^{-8} are used for numerical stability. The posterior latent state is appended to the history buffer and becomes available to the next causal prediction step [10,48].

The diagonal base process covariance $\mathbf{Q}_0$ is estimated from training-only latent prediction residuals, with a variance floor of 10^{-5}. The diagonal base measurement covariance $\mathbf{R}_0$ is estimated from the injected training-noise realization, with a variance floor of 10^{-6}. The applied covariances are

$$\mathbf{Q} = q\mathbf{Q}_0, \qquad \mathbf{R} = r\mathbf{R}_0, \tag{35}$$

where the scale factors are selected on validation-calibration from

$$q \in \{1, 2, 4, 8, 16\}, \qquad r \in \{0.5, 1, 2, 4\}. \tag{36}$$

The covariance-selection objective combines Gaussian negative log-likelihood with 0.25 times the mean prior RMSE.

3.8.8. Blended One-Step Prior

The reported one-step prior is not the unmodified LSTM output alone. It is a channel-wise convex combination of the raw LSTM/N4SID prior and the train-fitted VARX or persistence forecast:

$$\widehat{y}_{i,k}^- = \omega_i \widehat{y}_{i,k}^{\mathrm{LSTM},-} + (1 - \omega_i)\widehat{y}_{i,k}^{\mathrm{dyn}}, \qquad 0 \le \omega_i \le 1. \tag{37}$$

Each blend coefficient ω_i is fitted using clipped least squares on validation-calibration only. The blend modifies the reported output prior but does not replace the LSTM/N4SID measurement model used internally by the UKF latent-state update.

A global prediction-interval multiplier is also calibrated on validation-calibration from the 95th percentile of the absolute standardized innovations divided by 1.96. No test observation is used to select the blend weights, covariance scales, or uncertainty multiplier.

3.8.9. Five-Seed Full-Record Evaluation

After the sensitivity analysis, the selected warm-up and fine-tuning target shares of 0.20 and 0.50 are frozen. The complete PI-LSTM-UKF model is then trained independently on the full 100,000-sample record using the five predeclared random seeds.

For each trained realization, the base covariance matrices, covariance scale factors, interval multiplier, and prior-blend coefficients are calibrated independently using validation-calibration only. The complete validation interval is processed before test evaluation to obtain the causal filter state at the validation-test boundary. The untouched test block is then evaluated once for that realization.

No best test seed is selected. Final quantitative results are reported as the mean and standard deviation across the five independent full-record runs. Seed 11 is used only to provide a representative time-series visualization and does not determine the reported five-run statistics.

Test inference timing excludes training, validation calibration, plotting, file input/output, and initial execution compilation. Both the pre-measurement one-step prior and the measurement-assisted posterior are reported separately. This distinction prevents the filtering posterior from being incorrectly presented as autonomous forecasting performance.

3.9. Theoretical Justification of UKF Integration with Nonlinear LSTM

The unscented transform propagates moments through nonlinear mappings without requiring Jacobians and, under its standard Gaussian assumptions, retains third-order accuracy for the usual nonlinear expansion [55]. At inference, the LSTM is a smooth composition of affine mappings and sigmoid/tanh gates, while dropout is inactive. The physics-informed terms discourage transitions that are inconsistent with the training physics, although they do not by themselves guarantee stability. Stability is supported by the causal FIFO history, positive covariance floors, and Joseph-form update. Related recurrent-network/UKF estimators have been used in power, energy-storage, and robotic applications [56–58].

The PI-LSTM and UKF have distinct roles. The recurrent model learns the nonlinear pre-measurement dynamics, whereas the UKF propagates uncertainty and uses the current observation to correct the latent state. The ablation study in Section 5.2 evaluates these contributions separately.

3.10. Evaluation Protocol: Closed-Loop Filtering vs. Open-Loop Prediction

The comparison includes N4SID-state DMDc, delay-SINDy-c, DMDc-CNN, DMDc-LSTM, SINDy-CNN, and SINDy-LSTM. DMDc is evaluated as an open-loop latent rollout. Delay-SINDy-c is a one-step prior model, and the four residual hybrids are rolling one-step-ahead predictors that may use measurements through $k-1$, but never the current target $\mathbf{y}_k$. The evaluation mode and forecast horizon are reported for each method.

For the proposed model, $\widehat{\mathbf{y}}_k^-$ is calculated before assimilating $\mathbf{y}_k$ and is therefore used in predictive comparisons. The posterior $\widehat{\mathbf{y}}_k^+$ follows the UKF update with the current measurement and is reported separately as filtering performance. It is not ranked as an autonomous forecast. Prior and posterior MSE, and R^2 are reported, together with prior coverage and interval width.

The reported metrics are calculated in coordinates normalized by the training standard deviations. When the conditional power transformation is active, its normalized errors are correspondingly evaluated in log-transformed coordinates. Predictions are also inverse-transformed for engineering-unit plots, and any engineering-unit RMSE or MAE is calculated and labelled separately. All methods use the same test indices, preprocessing

statistics, and noisy measurement realization. The final configuration is repeated for the five seeds using `rng(seed,'twister')`.

These generator states control network initialization, mini-batch order, dropout, and scheduled-sampling masks. They have no physical interpretation and are fixed before test evaluation. No best seed is selected. Seed 11 is used only for representative plots, it does not affect aggregate results.

For a metric m_s, the paper reports

$$\overline{m} = \frac{1}{5}\sum_{s\in\mathcal{S}} m_s, \qquad \mathrm{SD}(m) = \sqrt{\frac{1}{4}\sum_{s\in\mathcal{S}}(m_s-\overline{m})^2}. \tag{38}$$

The same convention is used for inference time, milliseconds per sample, and throughput, timing excludes training, calibration, plotting, and file I/O. Because all runs share the same simulated record and noise realization, this standard deviation describes optimization variability rather than independent plant experiments. Model parameters are summarized in Table 4.

Table 4. PI-LSTM-UKF configuration used in the reported experiments.

Setting	Value
Record and split	100,000 samples at 1 s, 70/15/15 chronological
Validation use	7.5% validation-calibration, 7.5% val-select
Inputs and outputs	6 and 5
Sequence and LSTM sizes	30, 128 and 64 units
Warm-up/fine-tuning	5/at most 30 epochs
Batch/learning rate	$128/5\times10^{-4}$
Target shares	physics 0.20 warm-up and 0.50 fine-tuning
Bias/rollout	0.05/0.10, three-step horizon
UKF	$\alpha=0.1,\beta=2,\kappa=0$
Data/noise RNG	`rng(42,'twister')`
Reporting	five-run mean $\pm$ sample SD, no seed selection

4. Results

This section presents a systematic evaluation of the seven model variants under a common leakage-free protocol. The six baseline and residual-hybrid models are evaluated strictly as one-step-ahead (or open-loop) predictors that may use measurements only through time $k-1$. For the PI-LSTM-UKF two quantities are reported separately:

- the **one-step prior** $\widehat{\mathbf{y}}_k^-$, computed before the current measurement is assimilated;
- the **filtering posterior** $\widehat{\mathbf{y}}_k^+$, obtained after the UKF measurement update.

Only the one-step prior is used for predictive ranking against the open-loop baselines. The posterior characterises closed-loop filtering performance and is reported separately.

The seven model variants were assessed using high-fidelity Simscape data under realistic sensor noise ($\sigma=0.05$) and off-nominal cold-start conditions. Performance was measured using Mean Squared Error (MSE) and the coefficient of determination (R^2). Results are summarized in Tables 5–10. The evaluation focuses on assessing the contribution of each modeling component, linear identification, sparse knowledge extraction, residual learning, physics-informed regularization, and sequential state estimation, toward overall robustness and physical consistency.

Table 5. Baseline model performance (open-loop/one-step prediction). Mean squared error (MSE) and coefficient of determination (R^2) are reported for each output. The pure UKF is shown only for reference and is *not* part of the main seven-model hierarchy.

Output	UKF		DMDc		SINDy	
	MSE	R^2	**MSE**	R^2	**MSE**	R^2
V_{HP}	0.06084	0.8323	0.11570	0.6811	0.00599	0.9835
$\dot{Q}_{\text{FW pump}}$	0.24392	0.6951	0.21384	0.7327	0.03008	0.9624
$V_{\text{extraction}}$	0.05511	0.8704	0.20516	0.5175	0.01131	0.9734
η_{system}	0.11124	0.8513	0.50108	0.3302	0.01579	0.9789
P_{out}	0.07907	0.8432	0.26098	0.4825	0.00630	0.9875

Table 6. Mean Squared Error (MSE) comparison across different models (mean ± standard deviation over 5 independent runs). Lower values indicate better performance. Results are reported as mean ± standard deviation over five independent runs. (**open loop models**).

Output Variable	DMDc + CNN	SINDy + CNN	DMDc + LSTM	SINDy + LSTM
HP Turbine Valve	0.01184 ± 0.000933	0.00357 ± 0.0000933	0.00664 ± 0.000209	$2.675364 \times 10^{-3} \pm 4.77 \times 10^{-5}$
Feedwater Pump	0.01924 ± 0.000353	0.09594 ± 0.00334	0.01074 ± 0.000335	$7.534073 \times 10^{-3} \pm 7.13 \times 10^{-4}$
Steam Extraction Valve	0.02083 ± 0.000415	0.00743 ± 0.00167	0.01028 ± 0.000974	$6.254613 \times 10^{-3} \pm 4.73 \times 10^{-4}$
System Efficiency	0.01489 ± 0.000974	0.01195 ± 0.0000129	0.01156 ± 0.000415	$4.313063 \times 10^{-3} \pm 5.37 \times 10^{-5}$
Power (MW)	0.01074 ± 0.000429	0.00862 ± 0.0000185	0.00860 ± 0.0001788	$3.170371 \times 10^{-3} \pm 7.69 \times 10^{-5}$

Table 7. Coefficient of determination (R^2) comparison across different models (mean ± standard deviation over 5 independent runs). Higher values indicate better performance. Results are reported as mean ± standard deviation over five independent runs. (**open loop models**).

Output Variable	DMDc + CNN	SINDy + CNN	DMDc + LSTM	SINDy + LSTM
HP Turbine Valve	0.96736 ± 0.002573	0.990175 ± 0.000063	0.981692 ± 0.000576	0.992624 ± 0.000131
Feedwater Pump	0.975952 ± 0.000441	0.880071 ± 0.000715	0.986578 ± 0.000042	0.990583 ± 0.000892
Steam Extraction Valve	0.951019 ± 0.0093757	0.982535 ± 0.001189	0.975819 ± 0.003921	0.985290 ± 0.001112
System Efficiency	0.980093 ± 0.001302	0.984031 ± 0.000111	0.984542 ± 0.000173	0.994235 ± 0.000072
Power (MW)	0.978710 ± 0.00085	0.98290 ± 0.000126	0.982947 ± 0.000367	0.993714 ± 0.000152

For a predicted output $\hat{\mathbf{y}}(k)$ and true output vectors $\mathbf{y}_k$, the mean squared error for each output ($i = 1 \ldots 5$) and a number of samples ($k = N$) is:

$$\text{MSE}_i = \frac{1}{N} \sum_{k=1}^{N} (y_i(k) - \hat{y}_i(k))^2 \tag{39}$$

The coefficient of determination R^2 for each output ($i = 1 \ldots 5$) is defined as

$$R_i^2 = 1 - \frac{\sum_{k=1}^{N} (y_i(k) - \hat{y}_i(k))^2}{\sum_{k=1}^{N} (y_i(k) - \bar{y}_i)^2} \tag{40}$$

where $\bar{y}_i = \frac{1}{N} \sum_{k=1}^{N} y_i^{(k)}$ is the mean of the true values for output i.

Table 8. Training-standardized MSE for the PI-LSTM-UKF target physics-share sensitivity study under rolling closed-loop one-step-ahead prior prediction. At each test instant, the prior estimate was evaluated before assimilating the current measurement. The reported configurations specify the nominal target physics shares during the warm-up (s_{pre}) and fine-tuning (s_{fine}) stages. The effective physics-loss coefficients were computed automatically during training. Values are reported as mean ± standard deviation over five independent runs.

Output	PI-LSTM-UKF (0.00, 0.00) No Physics	PI-LSTM-UKF (0.10, 0.20) Low Share	PI-LSTM-UKF (0.20, 0.50) Selected	PI-LSTM-UKF (0.50, 0.20) Moderate Share	PI-LSTM-UKF (0.60, 0.50) High Share
HP Turbine Valve	$5.3257 \times 10^{-3} \pm 7.12 \times 10^{-4}$	$5.6860 \times 10^{-3} \pm 1.69 \times 10^{-3}$	$5.0226 \times 10^{-3} \pm 1.22 \times 10^{-4}$	$5.2444 \times 10^{-3} \pm 6.12 \times 10^{-4}$	$5.0422 \times 10^{-3} \pm 1.72 \times 10^{-4}$
Feedwater Pump	$4.5233 \times 10^{-3} \pm 2.17 \times 10^{-4}$	$4.4160 \times 10^{-3} \pm 1.69 \times 10^{-4}$	$4.3835 \times 10^{-3} \pm 7.93 \times 10^{-5}$	$4.3989 \times 10^{-3} \pm 8.81 \times 10^{-5}$	$4.5126 \times 10^{-3} \pm 1.11 \times 10^{-4}$
Steam Extraction Valve	$6.6446 \times 10^{-3} \pm 6.92 \times 10^{-4}$	$6.3323 \times 10^{-3} \pm 1.25 \times 10^{-4}$	$6.3498 \times 10^{-3} \pm 3.09 \times 10^{-4}$	$6.2874 \times 10^{-3} \pm 6.03 \times 10^{-5}$	$6.4625 \times 10^{-3} \pm 4.17 \times 10^{-4}$
System Efficiency	$2.5080 \times 10^{-3} \pm 2.16 \times 10^{-6}$	$2.5065 \times 10^{-3} \pm 3.54 \times 10^{-6}$	$2.5048 \times 10^{-3} \pm 3.23 \times 10^{-6}$	$2.5049 \times 10^{-3} \pm 3.22 \times 10^{-6}$	$2.5050 \times 10^{-3} \pm 3.26 \times 10^{-6}$
Power (MW)	$2.4687 \times 10^{-3} \pm 5.02 \times 10^{-9}$	$2.4690 \times 10^{-3} \pm 5.15 \times 10^{-7}$	$2.4687 \times 10^{-3} \pm 7.29 \times 10^{-9}$	$2.4689 \times 10^{-3} \pm 3.64 \times 10^{-7}$	$2.4688 \times 10^{-3} \pm 1.86 \times 10^{-7}$

Configuration (a, b) denotes the nominal target physics shares $(s_{\text{pre}}, s_{\text{fine}})$ used during warm-up and fine-tuning, respectively.

Table 9. Coefficient of determination (R^2) for the PI-LSTM-UKF target physics-share sensitivity study under rolling closed-loop one-step-ahead prior prediction. At each test instant, the one-step prior was evaluated before assimilating the current measurement. The reported values correspond to the nominal target physics shares used during training, whereas the effective physics-loss coefficient was calibrated automatically according to Equation (24). Results are reported as mean ± standard deviation over five independent runs.

Output	PI-LSTM-UKF (0.00, 0.00) No Physics	PI-LSTM-UKF (0.10, 0.20) Low Share	PI-LSTM-UKF (0.20, 0.50) Selected	PI-LSTM-UKF (0.50, 0.20) Moderate Share	PI-LSTM-UKF (0.60, 0.50) High Share
HP Turbine Valve	0.985318 ± 0.001963	0.984325 ± 0.004664	0.986153 ± 0.000337	0.985542 ± 0.001687	0.986099 ± 0.000475
Feedwater Pump	0.994346 ± 0.000271	0.994480 ± 0.000212	0.994521 ± 0.000099	0.994502 ± 0.000110	0.994359 ± 0.000139
Steam Extraction Valve	0.984373 ± 0.001628	0.985107 ± 0.000294	0.985066 ± 0.000727	0.985213 ± 0.000142	0.984801 ± 0.000981
System Efficiency	0.996648 ± 0.000003	0.996650 ± 0.000005	0.996652 ± 0.000004	0.996652 ± 0.000004	0.996652 ± 0.000004
Power (MW)	$0.995105 \pm 9.95 \times 10^{-9}$	$0.995105 \pm 1.02 \times 10^{-6}$	$0.995105 \pm 1.45 \times 10^{-8}$	$0.995105 \pm 7.21 \times 10^{-7}$	$0.995105 \pm 3.69 \times 10^{-7}$

Configuration (a, b) denotes the nominal target physics shares $(s_{\text{pre}}, s_{\text{fine}})$ used during warm-up and fine-tuning, respectively.

Table 10. MSE and coefficient of determination (R^2) results for the ablation comparison of LSTM alone, PI-LSTM only, and closed-loop PI-NN-UKF models.

Output	LSTM Alone		PI-LSTM Only		PI-NN + UKF	
	MSE	R^2	**MSE**	R^2	**MSE**	R^2
HP Turbine Valve	0.0548	0.8490	0.0062	0.9830	0.0054	0.9852
Feedwater Pump	0.3000	0.6250	0.0126	0.9842	0.0103	0.9871
Steam Extraction Valve	0.0208	0.9510	0.0120	0.9718	0.0087	0.9795
System Efficiency	0.1436	0.8080	0.1335	0.8215	0.0524	0.9299
Power (MW)	0.0817	0.8380	0.0651	0.8710	0.0557	0.8895

4.1. Baseline Models

The N4SID-state DMDc and delay-SINDy-c models were evaluated as open-loop/one-step-ahead predictors under the common leakage-free protocol. SINDy provided the stronger analytical baseline, attaining R^2 values in the range 0.9624–0.9875 and MSE values ranging from 0.00599 to 0.03008 across the five outputs (Figures 5–7). DMDc captured the dominant linear dynamics but showed higher errors on several channels and was less accurate on fast transients. Both models therefore leave residual errors that motivate the subsequent CNN- and LSTM-based residual-learning stages. No Unscented Kalman Filter is applied to these baselines; the UKF is used only inside the final PI-LSTM-UKF architecture.

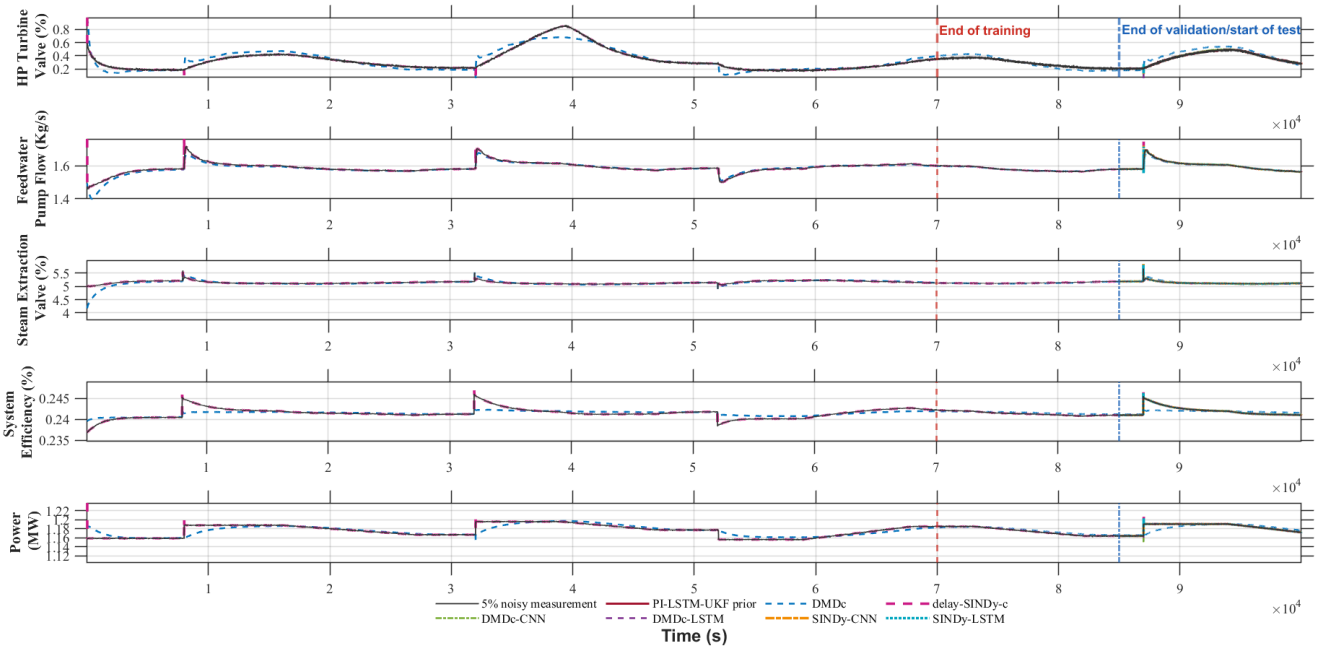


Figure 5. Comparison of the measured outputs and the one-step-ahead predictions generated by all benchmark and proposed hybrid models over the complete 100,000-sample dataset. The full-record visualization illustrates the overall prediction accuracy and dynamic tracking performance of each model across the chronological training, validation, and test partitions.

4.2. Performance of the Proposed CNN-Based Residual Correction

Residual learning using 1D CNNs was applied to compensate for localized nonlinear patterns not captured by the analytical baselines. The DMDc + CNN residual model substantially reduced MSE relative to the DMDc baseline, with reductions of approximately 90–97% across the five outputs. Relative to the stronger SINDy baseline, however, the effect of CNN residual correction was mixed across outputs (Tables 6 and 7). SINDy + CNN achieved R^2 values ranging from approximately 0.88 to 0.99, indicating that sparse

identification combined with localized residual learning can provide an effective intermediate solution. Nevertheless, the CNN architecture remains more limited in representing long-range thermal dependencies.

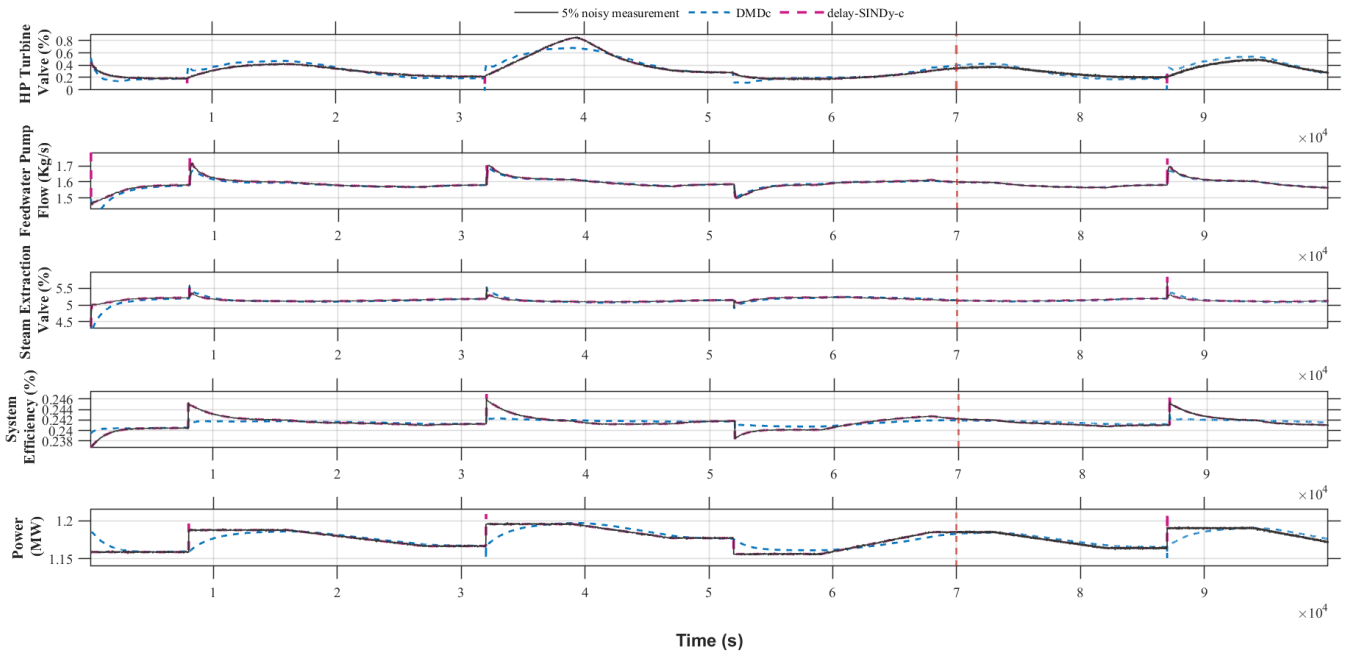


Figure 6. Comparison of the measured outputs and the predictions generated by the baseline analytical models (N4SID-state DMDc and delay-SINDy-c) over the complete test sequence.

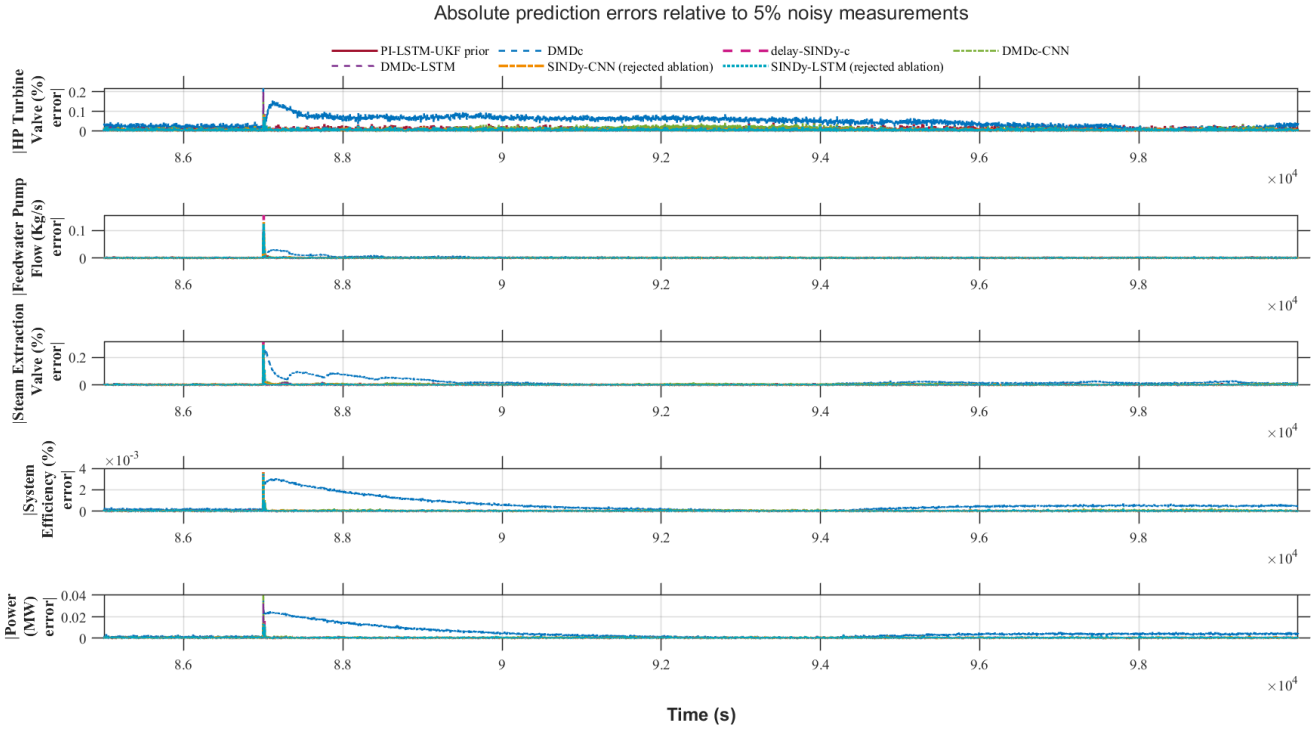


Figure 7. Zoomed-in views of the absolute prediction errors of the benchmark and hybrid models over representative intervals of the held-out test set. The enlarged plots highlight local differences in prediction accuracy under noisy measurements and facilitate comparison of the error characteristics among the evaluated models.

4.3. Performance of the Proposed LSTM-Based Residual Correction

Replacing CNNs with LSTM networks for residual correction further improved performance, particularly for variables influenced by delayed thermal effects. The SINDy + LSTM configuration reduced MSE relative to SINDy + CNN on all five outputs and achieved $R^2 > 0.98$ across all monitored variables (Tables 6 and 7). These results demonstrate the advantage of recurrent architectures in representing long-range temporal dependencies associated with thermal inertia and delayed energy transport.

Overall, LSTM-based residual correction outperforms the CNN-based counterpart for critical variables such as feedwater pump flow, system efficiency, and electrical power output.

These results highlight the value of residual learning, however, purely data-driven LSTM models remain vulnerable to physical inconsistency during extrapolation, motivating the integration of explicit thermodynamic constraints.

4.4. Proposed PI-LSTM-UKF

Training of the PI-LSTM-UKF follows a two-stage supervised protocol. A short physics-weighted warm-up (5 epochs) is followed by fine-tuning (at most 30 epochs) with early stopping on the validation-selection partition. Both stages contain supervised latent-state, output, bias and rollout terms together with the composite physics regulariser. Physics regularisation is controlled by *target shares* rather than a fixed raw coefficient λ_{phys}. The selected shares are

$$s_{\text{pre}} = 0.20 \quad \text{(warm-up)}, \qquad s_{\text{fine}} = 0.50 \quad \text{(fine-tuning)}.$$

At the beginning of each stage the corresponding effective coefficient is recalibrated from the current magnitudes of the data and physics losses according to

$$\lambda_{\text{phys}}(s) = \frac{s}{1-s} \frac{\mathcal{L}_{\text{cal}}}{\max(\mathcal{L}_{\text{phys}}, \epsilon)}.$$

Consequently the numerical values 0.20 and 0.50 are target shares, not direct values of λ_{phys}. The PI-LSTM-UKF achieved high one-step-prior accuracy and strong closed-loop filtering performance.

As shown in Tables 8 and 9, the PI-LSTM-UKF prior provides improved generalization and physical consistency, with $R^2 \geq 0.984$ across all outputs. The corresponding posterior estimates (Table 11) achieve $R^2 = 1.0000$, while MSE values range between 10^{-6} and 10^{-7}. Figure 5 presents the full-record comparison, while Figure 8 provides a detailed view of a representative time-series segment. The complete power-output trajectory and three representative magnified regions are shown in Figure 10, enabling a detailed comparison of the local tracking performance of the evaluated models.

For clarity, two evaluation modes are distinguished. DMDc is evaluated using open-loop rollout, whereas delay-SINDy-c and the CNN or LSTM residual hybrids are evaluated as causal rolling one-step-ahead predictors using information available only through $k-1$. The full PI-LSTM-UKF is evaluated in closed-loop filtering mode, where the UKF performs a measurement update using the current noisy observation. Therefore, the main PI-LSTM-UKF results should be interpreted as filtered state estimation rather than pure open-loop forecasts. To ensure a fair comparison, the PI-LSTM is evaluated separately in open-loop mode without state estimation, while the PI-LSTM-UKF is evaluated in both one-step-ahead prior (prediction before measurement assimilation) and closed-loop posterior (prediction followed by measurement update) modes. The posterior PI-LSTM-UKF is

compared directly with a conventional UKF baseline using an identical measurement-update mechanism.

Table 11. Posterior filtering performance of the final PI-LSTM-UKF model obtained using the selected target physics-share configuration $(s_{\text{pre}}, s_{\text{fine}}) = (0.20, 0.50)$ after assimilation of the current test measurement. The MSE values are reported in training-standardized coordinates.

Output	Standardized Posterior MSE	R^2
HP turbine valve	1.2951×10^{-6}	1.0000
Feedwater pump	2.5968×10^{-6}	1.0000
Steam extraction valve	3.2824×10^{-7}	1.0000
System efficiency	2.4908×10^{-7}	1.0000
Generated power	1.0751×10^{-6}	1.0000

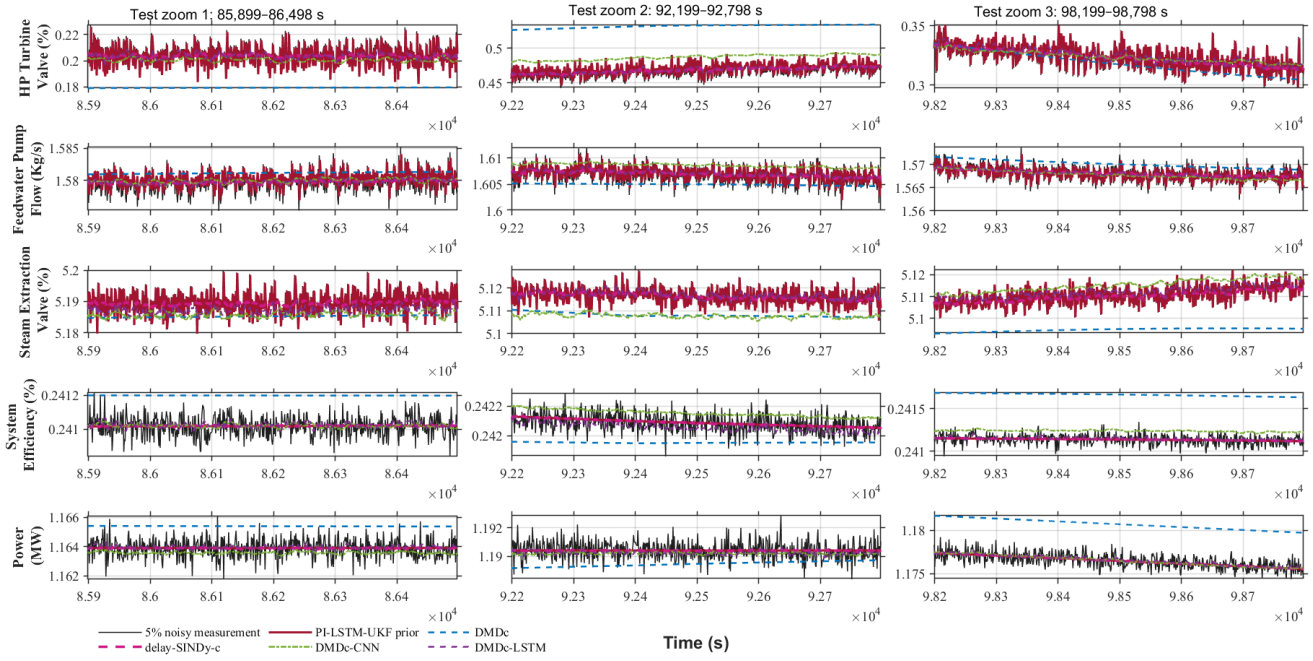


Figure 8. Zoomed-in views of the signed prediction errors for all benchmark and proposed hybrid models over representative regions of the complete dataset. Positive and negative prediction deviations are preserved to facilitate comparison of local prediction bias and tracking performance.

UKF Posterior Filtering

The one-step-ahead prior estimates were first obtained using the UKF prediction (time-update) stage, in which the learned PI-LSTM process model propagated the sigma points without incorporating the current output measurement. Subsequently, the UKF assimilated the current measurement through its correction (measurement-update) stage to generate the posterior state estimate. The standardized posterior filtering results are reported in Table 11. The posterior estimates achieved standardized MSE values between 10^{-7} and 10^{-6}, while the corresponding R^2 values rounded to 1.0000 at four decimal places for all five outputs.

The near-unity posterior coefficients of determination demonstrate that the UKF effectively refines the prior state estimate by incorporating the current measurement. However, these posterior results characterize state estimation and measurement reconstruction rather than independent forecasting because the current output measurement directly participates in the correction step. Consequently, the posterior results are reported separately and are not used for predictive comparisons with open-loop recurrent models or one-step-ahead prior predictions that do not assimilate the current measurement.

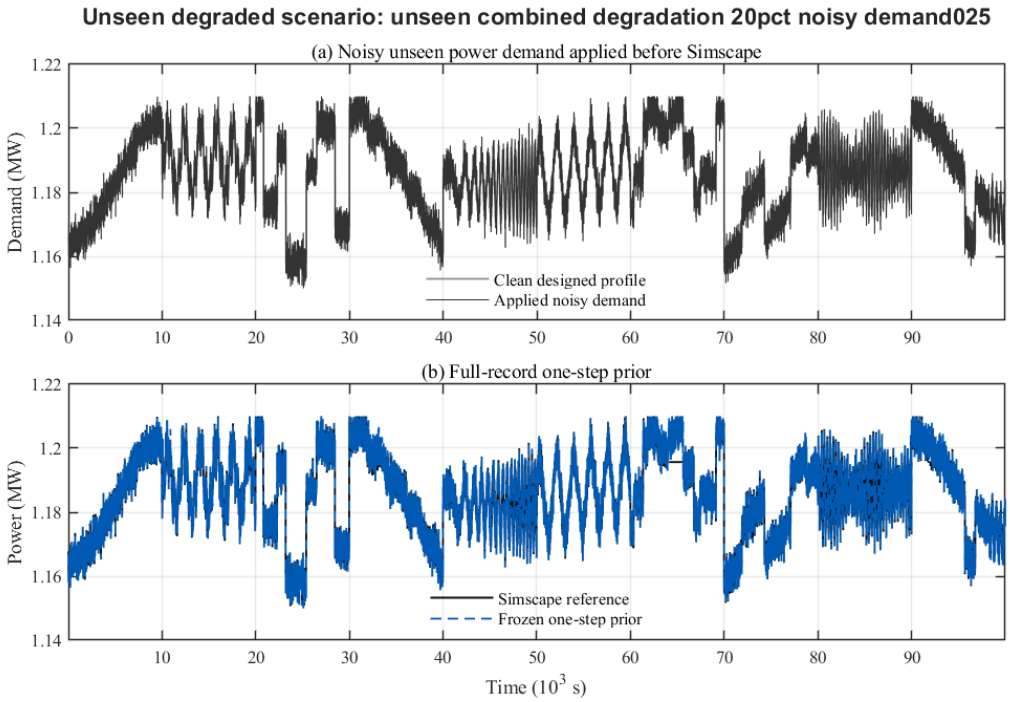


Figure 9. Full-record evaluation of the frozen PI-LSTM-UKF under the independent unseen scenario. The test combines a nonrepeating 100,000 s demand profile, simultaneous 20% degradation of the turbine efficiencies and heat-transfer coefficients, and additional Gaussian measurement noise with a normalized standard deviation of 0.25. Panel (**a**) presents the applied power-demand trajectory, while panel (**b**) compares the Simscape reference with the frozen one-step prior and the measurement-assisted UKF posterior. The frozen prior follows the degraded-plant power response across ramps, steps, oscillations, chirp-like excitation, and load-rejection events without retraining, refitting, or recalibration.

The normalized standard deviations of the test outputs were 0.999, 0.964, 0.958, 0.987, and 0.998 for the HP turbine valve, feedwater pump, steam extraction valve, system efficiency, and generated power, respectively. These values indicate that the test set retained substantial output variability relative to the training set and that the reported performance was not obtained from an approximately constant operating segment.

4.5. Ablation Study

Ablation experiments (Tables 8–10) further clarify the contribution of individual components. As mentioned earlier the sliding windows ($w = 30$) are split (70%/15%/15%) [35]. Removing the UKF reduced R^2 by 0.0103 for the feedwater pump, 0.1752 for system efficiency, and 0.1241 for power, while the corresponding MSE values increased by factors of approximately 3.4, 56.0, and 37.5, respectively. Replacing the recurrent LSTM with a feedforward neural network also reduced performance, with R^2 decreases of 0.0668 for system efficiency and 0.1056 for power. Sensitivity analysis with respect to the target physics shares s_{pre} and s_{fine} shows selected shares $s_{\text{pre}} = 0.2$ and $s_{\text{fine}} = 0.5$, where the model achieves a good balance between data-driven accuracy and physical consistency while maintaining low variance across multiple random seeds.

4.6. Critical Interpretation of Performance Metrics

The high R^2 values (generally ≥ 0.984, reaching 0.995 on some outputs) and low MSE achieved by the PI-LSTM-UKF framework must be interpreted with caution. These results were obtained on a high-fidelity but idealized Simscape simulation with relatively smooth dynamics, bounded Gaussian noise ($\sigma = 0.05$), and normalized output variables. The

strong alignment between the embedded thermodynamic constraints and the underlying Simscape model assumptions also contributes to the observed performance. While these results demonstrate the effectiveness of the proposed hybrid architecture under controlled conditions, they represent an upper-bound benchmark. Performance is expected to degrade on real plant data, where sensor drift, unmodeled hysteresis, fouling, and varying degradation introduce additional challenges. This limitation is further examined in Section 5.

4.7. Unseen Load Profile Under Degraded and Noisy Conditions

The frozen PI-LSTM-UKF was evaluated on an independent 100,000 s load profile sampled at 1 s. The profile contained previously unseen ramps, irregular steps, sinusoidal segments, chirp-like excitation, triangular cycles, random dwell levels, and temporary load-rejection events.

A simultaneous 20% component degradation was imposed. The HP and LP turbine isentropic and mechanical efficiencies were reduced from 0.90 to 0.72, while the boiler, preheater, and condenser heat-transfer coefficients were reduced from 200, 12,000, and 3500 to 160, 9600, and 2800 $\mathrm{W\,m^{-2}K^{-1}}$, respectively. Gaussian measurement noise with normalized standard deviation 0.25 was added. No bias, drift, outliers, dropouts, or external filtering were applied. All metrics were calculated against the clean Simscape references.

The complete prediction pipeline remained frozen: no network retraining, N4SID or VARX refitting, normalization adjustment, covariance recalibration, or prior-blend recalibration was performed. Figure 9 shows that the one-step prior remained stable throughout the complete unseen record. The enlarged responses in Figure 11 demonstrate accurate tracking during ramps, steps, chirp-like excitation, and amplitude-modulated operation. Figure 12 further confirms that the frozen predictor reproduced the changing-frequency dynamics of all five outputs.

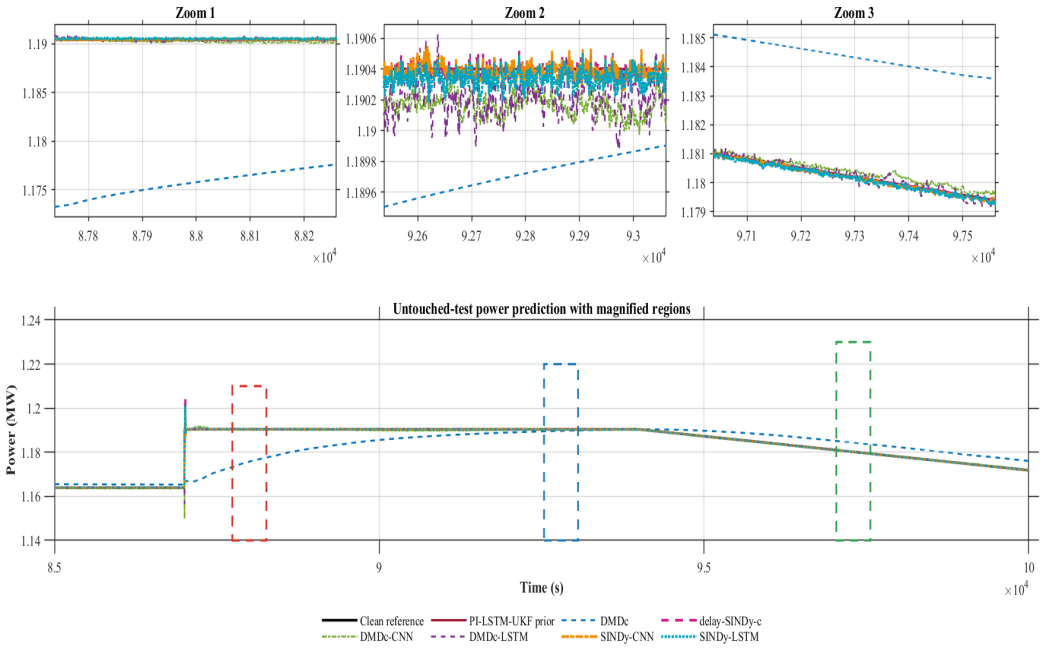


Figure 10. Zoomed-in views of three representative regions from the complete dataset showing the signed prediction errors (prediction—measured output) of all benchmark and proposed hybrid models. The enlarged intervals facilitate detailed comparison of local tracking accuracy and prediction bias under noisy measurements.

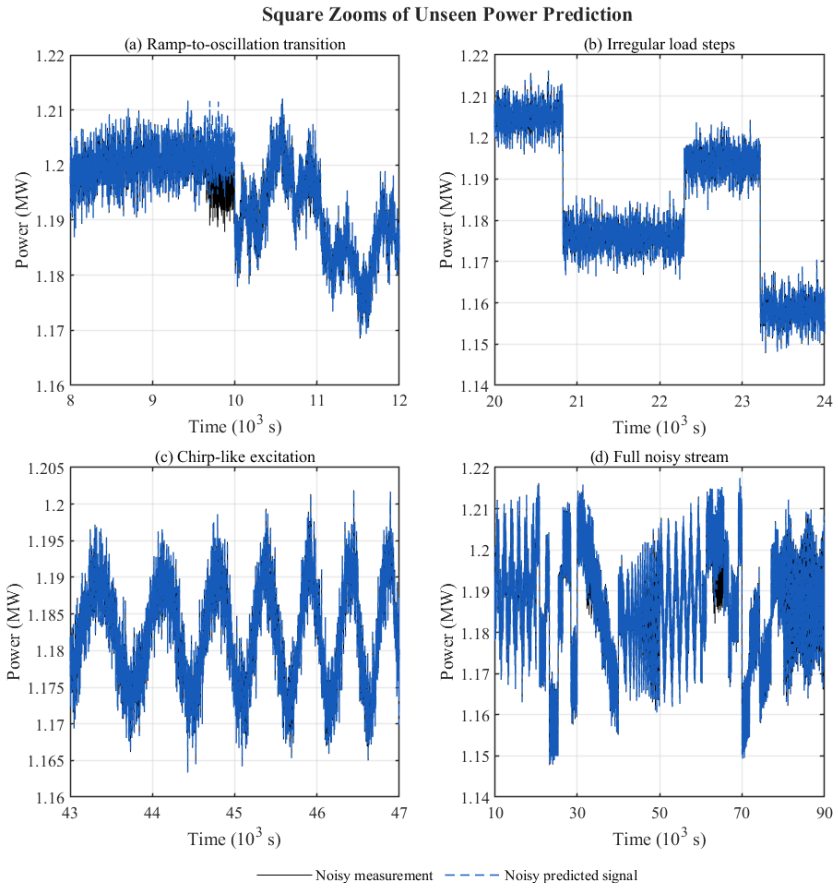


Figure 11. Enlarged generated-power responses during four representative portions of the independent unseen scenario: (**a**) a ramp-to-oscillation transition, (**b**) irregular load steps, (**c**) chirp-like excitation, and (**d**) amplitude-modulated excitation. The frozen one-step prior remains closely aligned with the Simscape reference during both smooth and abrupt dynamic changes. The UKF posterior is visibly more variable because it incorporates measurements corrupted by normalized Gaussian noise with standard deviation 0.25 while retaining the frozen validation-calibrated measurement covariance.

As reported in Table 12, the prior achieved R^2 values of 0.9693, 0.9766, 0.8493, 0.9621, and 0.9851 for the HP turbine valve, feedwater pump, extraction valve, system efficiency, and generated power, respectively. The mean prior R^2 was 0.9485.

The results indicate good generalization of the frozen one-step predictor under a dynamically diverse unseen load, simultaneous component degradation, and increased measurement noise. Nevertheless, validation using operational plant measurements remains necessary before extending these conclusions to industrial deployment.

Table 12. Frozen one-step-prior performance under the independent unseen load profile with 20% combined component degradation and additional normalized measurement-noise standard deviation of 0.25. Metrics are evaluated against the clean Simscape reference in actual output units.

Output Variable	MSE	RMSE	MAE	R^2
HP Turbine Valve	1.4299×10^{-9}	3.7814×10^{-5}	3.0144×10^{-5}	0.9693
Feedwater Pump	3.8045×10^{-5}	0.006168	0.004893	0.9766
Steam Extraction Valve	3.1659×10^{-14}	1.7793×10^{-7}	1.3913×10^{-7}	0.8493
System Efficiency	8.8001×10^{-8}	2.9665×10^{-4}	7.5691×10^{-5}	0.9621
Power (MW)	2.0886×10^{-6}	0.001445	0.000336	0.9851

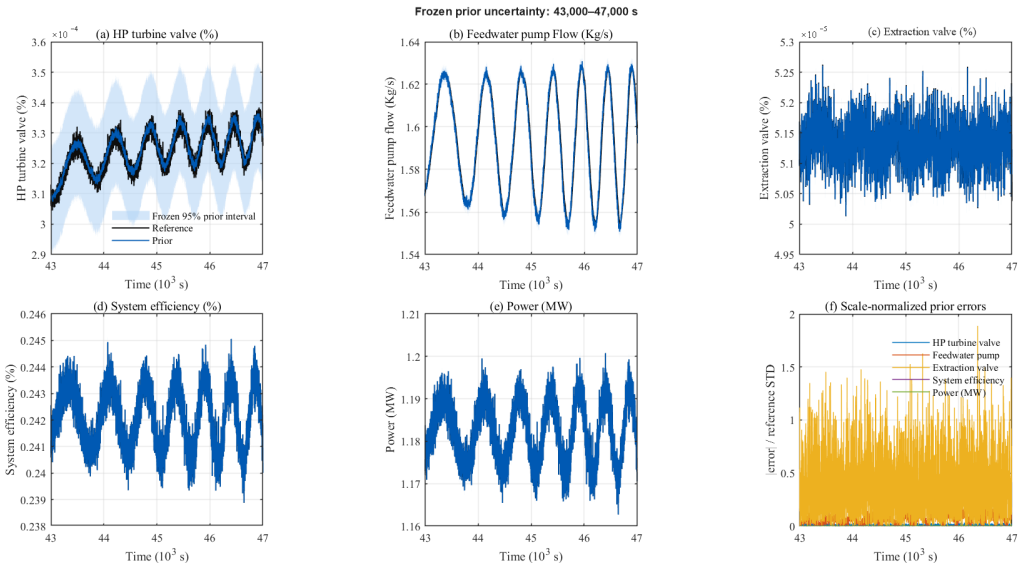


Figure 12. Multi-output response over the 43,000-47,000 s chirp-like excitation interval. Panels show (**a**) HP turbine-valve response, (**b**) feedwater-pump flow, (**c**) extraction-valve response, (**d**) system efficiency, (**e**) generated power, and (**f**) applied power demand. The frozen PI-LSTM-UKF prior reproduces the changing-frequency dynamics of the degraded Simscape plant across all five outputs. The comparatively larger relative dispersion of the extraction-valve channel is consistent with its lower prior coefficient of determination, whereas generated power and system efficiency retain strong agreement with the reference.

5. Discussion

The progressive improvement observed across the evaluated models reflects the complementary roles of their constituent components rather than the effect of a single algorithmic modification. Each successive stage addresses a specific limitation of the preceding one. Linear system-identification models provide an interpretable approximation of the dominant dynamics but cannot fully represent the nonlinear thermodynamic interactions of the Rankine cycle. Residual neural networks, implemented using CNN and LSTM architectures, compensate for these remaining nonlinearities, physics-informed training constrains the learned dynamics toward thermodynamic consistency, and recursive Bayesian filtering reduces the influence of measurement uncertainty through sequential state estimation. The resulting hierarchy therefore illustrates how combining domain knowledge with data-driven learning and probabilistic estimation produces progressively more accurate and physically consistent digital twins.

5.1. Analysis of the Results

The analytical DMDc and SINDy baselines establish the performance achievable using explicit system-identification techniques without deep learning (Table 5). DMDc approximates the plant using a linear reduced-order state-space representation and therefore captures only the dominant operating dynamics, leading to lower predictive accuracy, particularly for outputs governed by strongly nonlinear thermodynamic interactions. Its reduced flexibility becomes particularly apparent for strongly coupled thermodynamic variables, where nonlinear interactions and delayed responses cannot be represented adequately by a fixed linear model. In contrast, SINDy benefits from sparse nonlinear basis functions, allowing it to retain a compact analytical structure while capturing part of the underlying nonlinear behavior. It achieved high accuracy across all outputs, with R^2 values ranging from 0.9624 (feedwater pump) to 0.9875 (generated power). This explains its consistently higher predictive accuracy relative to DMDc across all outputs.

Introducing CNN- and LSTM-based residual learning produced substantial but model-dependent changes in prediction accuracy (Tables 6 and 7). DMDc + CNN markedly improved upon the linear DMDc baseline across all outputs, whereas CNN correction of the stronger SINDy baseline produced mixed results across channels. The neural network therefore does not learn the complete plant dynamics, but instead models the remaining structured prediction error after the analytical approximation. This residual-learning strategy reduces the complexity of the learning task and enables the CNN or LSTM to focus on localized nonlinear corrections. CNN models reduce MSE compared to linear hybrids, with SINDy + CNN achieving $R^2 = 0.88$–0.99. Nevertheless, the convolutional architecture primarily exploits short-range temporal patterns through finite receptive fields. Consequently, although CNN residual correction reduces prediction errors considerably, its ability to model delayed thermal interactions extending over longer time horizons remains limited.

Replacing the convolutional residual corrector with an LSTM further improves prediction because recurrent memory enables the network to preserve information over substantially longer temporal horizons. Consistent with this interpretation, the LSTM residual corrector provided an additional reduction in MSE while increasing the SINDy-hybrid R^2 beyond 0.98 for all outputs. Thermal power systems exhibit delayed energy transport, thermal inertia, and coupled transient responses whose evolution depends on previous operating states rather than only recent observations. The recurrent architecture therefore models these long-term dependencies more effectively than localized convolutional filters, explaining the consistent improvement observed across most outputs despite a 32% increase in computational cost.

The additional improvement obtained by the proposed PI-LSTM-UKF does not arise solely from increasing model complexity. Instead, the embedded thermodynamic constraints regularize the optimization process by discouraging physically implausible trajectories during training. Consequently, the recurrent network is guided toward latent representations that simultaneously minimize prediction error and satisfy the imposed conservation relationships. This additional physical guidance becomes particularly beneficial under noisy measurements and operating conditions that differ from those encountered during training, where purely data-driven models are more susceptible to overfitting and extrapolation errors.

The full PI-LSTM-UKF prior consistently achieved $R^2 \geq 0.984$ across all five outputs. By embedding thermodynamic constraints through a two-stage training strategy and coupling the learned recurrent dynamics with UKF-based recursive state estimation, the model demonstrates substantially improved generalization and physical consistency. The proposed PI-LSTM-UKF additionally provides recursive Bayesian state estimation through the UKF, enabling measurement-assisted posterior filtering and uncertainty quantification, capabilities that are not available in the open-loop SINDy + LSTM model.

Unlike the embedded thermodynamic constraints, which improve the learned process model during training, the UKF contributes during inference by recursively propagating the predictive covariance and assimilating incoming measurements. Consequently, the two components play complementary roles: physics-informed learning improves the quality of the one-step prior, whereas recursive Bayesian filtering refines the online state estimate and provides uncertainty quantification. This complementary behavior explains the overall performance improvements achieved by the complete PI-LSTM-UKF framework in the ablation study. Removing the UKF reduced R^2 by approximately 0.175 for system efficiency and 0.124 for power, while replacing the recurrent LSTM with a feedforward network reduced R^2 by approximately 0.067 and 0.106 for the same outputs, respectively.

Although the PI-LSTM-UKF incurs additional computational cost because of recursive UKF propagation, the two-stage training strategy converged within approximately 5–15 epochs through early stopping, compared with approximately 20 epochs for the residual-learning baselines. Furthermore, the validation-only sensitivity analysis identified the target-share configuration $(s_{\text{pre}}, s_{\text{fine}}) = (0.20, 0.50)$ as a balanced operating point between predictive accuracy and thermodynamic regularization. This configuration was frozen before the final five-seed evaluation to prevent test-set driven hyperparameter selection.

5.2. Uncertainty Quantification and Calibration

The proposed PI-LSTM-UKF framework provides uncertainty quantification in addition to point predictions. At each time step, the Unscented Kalman Filter (UKF) propagates the prior state covariance through the nonlinear transition model. The resulting output covariance is used to construct one-step-ahead prediction intervals before assimilation of the current measurement.

The covariance-scaling parameters were estimated exclusively using the validation-calibration subset and were frozen before evaluation on the chronologically held-out test set. The 3σ prediction interval for each output was constructed as

$$\left[\hat{y}_{k|k-1} - 3\sigma_{k|k-1},\ \hat{y}_{k|k-1} + 3\sigma_{k|k-1}\right], \tag{41}$$

where $\hat{y}_{k|k-1}$ is the one-step-ahead prior prediction and $\sigma_{k|k-1}$ is the corresponding predicted output standard deviation. Under an ideal Gaussian assumption, these bounds have a nominal coverage of approximately 99.73%.

Uncertainty quality was evaluated using two complementary metrics: empirical prediction-interval coverage probability and mean prediction-interval width (MPIW). Coverage measures the percentage of held-out observations contained within the prediction intervals, whereas MPIW quantifies interval sharpness and allows the reported coverage to be interpreted alongside interval width.

The evaluation was repeated using five predetermined neural network initialization seeds while retaining the same simulated record, noise realization, chronological training-validation-test partitions, preprocessing parameters, and evaluation protocol. Therefore, the reported standard deviations quantify variability associated with stochastic initialization and minibatch ordering rather than robustness across independent operating scenarios.

As shown in Table 13, the mean empirical coverage across the five output summaries was 99.108%, compared with the nominal Gaussian coverage of approximately 99.73%. The steam extraction valve, system efficiency, and power achieved coverage close to or slightly above the nominal level. The HP turbine valve and feedwater pump achieved lower coverage of 98.380% and 97.239%, respectively, indicating channel-dependent undercoverage.

Table 13. Empirical coverage and mean width of the 3σ one-step-ahead prediction intervals on the untouched test set. Values are reported as mean ± standard deviation across five random initialization seeds. The nominal Gaussian coverage is approximately 99.73%.

Output Variable	Coverage (%)	Normalized MPIW	Engineering-Unit MPIW
HP Turbine Valve	98.380 ± 0.980	0.35014 ± 0.01157	$(5.4988 \pm 0.1817) \times 10^{-5}$
Feedwater Pump	97.239 ± 0.396	0.28771 ± 0.00639	0.008930 ± 0.000198
Steam Extraction Valve	99.935 ± 0.003	0.69268 ± 0.03409	$(4.8398 \pm 0.2382) \times 10^{-7}$
System Efficiency	100.000 ± 0.000	0.42900 ± 0.00651	$0.00050459 \pm 0.00000765$
Power (MW)	99.987 ± 0.005	0.40041 ± 0.00790	0.005274 ± 0.000104
Mean across outputs	99.108	0.43199	–

The interval-width results allow the empirical coverage values to be interpreted alongside interval sharpness. In normalized units, the steam extraction valve produced the widest intervals, with an MPIW of 0.69268 ± 0.03409, whereas the feedwater pump produced the narrowest intervals, with an MPIW of 0.28771 ± 0.00639. Thus, the overall mean coverage should not be interpreted as uniformly ideal calibration across every output. Rather, the results indicate strong aggregate coverage with remaining channel-dependent differences in calibration and sharpness.

Figure 13 presents the one-step-ahead prior predictions and corresponding 3σ prediction intervals for representative seed 11 on the untouched test set. The figure is provided as a representative visualization, while the quantitative results in Table 13 are aggregated across all five initialization seeds.

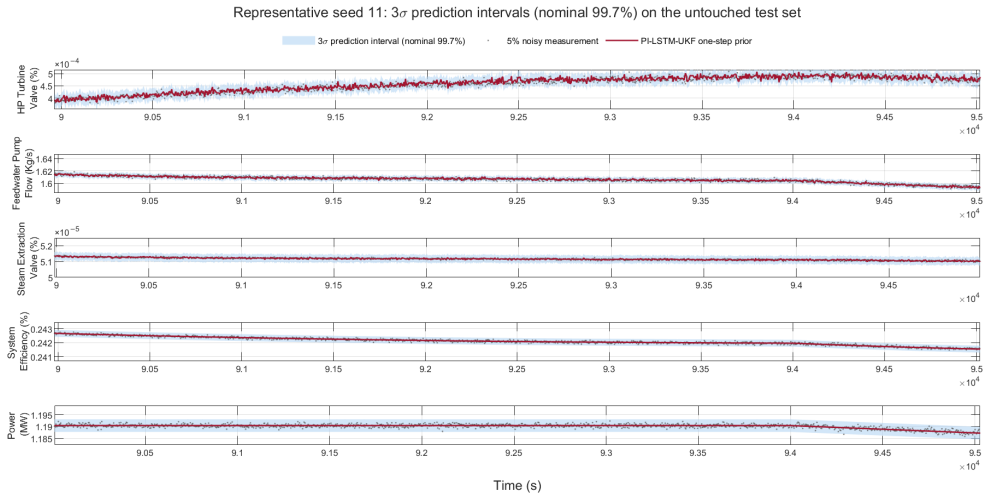


Figure 13. Representative seed 11 one-step-ahead predictions and corresponding 3σ prediction intervals on the untouched test set. The shaded regions represent $\hat{y}_{k|k-1} \pm 3\sigma_{k|k-1}$, with nominal coverage of approximately 99.73% under a Gaussian assumption. The points represent measurements containing Gaussian noise with a standard deviation equal to 5% of the training-signal standard deviation, and the red curves represent the PI-LSTM-UKF one-step-ahead prior predictions.

Throughout this section, the term *prediction interval* is used because the bounds quantify uncertainty in one-step-ahead predicted outputs. They are not confidence intervals for estimated population means.

Contribution of the Physics-Informed LSTM Component

To assess the contribution of each constituent module, a controlled ablation study was conducted by progressively constructing the proposed framework from four intermediate configurations. Table 14 reports the prediction performance of (i) a Pure UKF employing only the linear N4SID state-transition model, (ii) a data-driven LSTM without physics-informed regularization or recursive filtering, (iii) an open-loop Physics-Informed LSTM (PI-LSTM), and (iv) a closed-loop PI-NN-UKF, in which the recurrent predictor is replaced by a feedforward neural network. These intermediate configurations are subsequently compared with the complete PI-LSTM-UKF framework in Table 15.

The ablation results highlight the complementary contributions of the individual components. The Pure UKF provides stable recursive state estimation but is fundamentally constrained by its linear process model, resulting in comparatively lower predictive accuracy across several outputs. For example, the feedwater pump achieves an R^2 of only 0.6951, while the remaining variables exhibit coefficients of determination ranging from 0.8323 to 0.8704.

Table 14. Controlled ablation study: Normalized Mean Squared Error (MSE) and coefficient of determination (R^2) results for the ablation comparison of UKF, LSTM alone, PI-LSTM only, and closed-loop PI-NN-UKF models.

Output	UKF Alone		LSTM Alone		PI-LSTM Only		PI-NN + UKF	
	MSE	R^2	MSE	R^2	MSE	R^2	MSE	R^2
HP Turbine Valve	0.0608	0.8323	0.0548	0.8490	0.0062	0.9830	0.0054	0.9852
Feedwater Pump	0.2439	0.6951	0.3000	0.6250	0.0126	0.9842	0.0103	0.9871
Steam Extraction Valve	0.0551	0.8704	0.0208	0.9510	0.0120	0.9718	0.0087	0.9795
System Efficiency	0.1112	0.8513	0.1436	0.8080	0.1335	0.8215	0.0524	0.9299
Power (MW)	0.0791	0.8432	0.0817	0.8380	0.0651	0.8710	0.0557	0.8895

Table 15. Comparison between the Pure UKF baseline and the proposed PI-LSTM-UKF prior on the independent test set.

Output Variable	Pure UKF R^2	PI-LSTM-UKF Prior R^2	ΔR^2
HP Turbine Valve	0.8323	0.986358	+0.1541
Feedwater Pump	0.6951	0.994474	+0.2994
Steam Extraction Valve	0.8704	0.985407	+0.1150
System Efficiency	0.8513	0.996651	+0.1454
Power (MW)	0.8432	0.995105	+0.1519
Mean across outputs	0.8185	0.991599	+0.1731

Replacing the linear transition model with a conventional LSTM enables the network to capture nonlinear temporal relationships directly from the data. This leads to improved prediction for some variables, particularly the steam extraction valve, whose coefficient of determination increases from 0.8704 to 0.9510. However, the purely data-driven LSTM exhibits inconsistent behavior across the remaining outputs, with noticeable degradation for the feedwater pump and system efficiency. These results suggest that learning system dynamics solely from data does not provide sufficient physical guidance for reliable generalization.

Incorporating physics-informed regularization substantially enhances the open-loop recurrent model. The PI-LSTM achieves excellent predictive accuracy for the HP turbine valve and feedwater pump, reaching R^2 values of 0.9830 and 0.9842, respectively, while maintaining consistently strong performance for the remaining outputs. These improvements indicate that embedding thermodynamic constraints during training guides the network toward physically consistent representations without compromising its ability to model nonlinear dynamics.

The role of recurrent temporal modeling is assessed by comparing the PI-NN-UKF configuration with the complete PI-LSTM-UKF framework. Although PI-NN-UKF benefits from physics-informed training and recursive Bayesian filtering, its performance remains below that of the complete recurrent framework. For example, PI-NN-UKF achieves $R^2 = 0.9871$, 0.9299, and 0.8895 for the feedwater pump, system efficiency, and power outputs, respectively, whereas the corresponding PI-LSTM-UKF prior values are 0.9945, 0.9967, and 0.9951. This matched comparison indicates that recurrent temporal modeling provides an important contribution beyond physics-informed regularization and UKF correction.

The overall contribution of the proposed hybrid framework is quantified in Table 15, where the complete PI-LSTM-UKF is directly compared with the Pure UKF baseline. Replacing the linear process model with the Physics-Informed LSTM while retaining recursive Bayesian state estimation yields substantial improvements for every monitored variable. The consistently lower prediction errors and higher coefficients of determination demonstrate that neither recursive filtering nor physics-informed learning alone is sufficient to

achieve the best performance. Instead, the superior accuracy of the proposed framework arises from the synergy between physics-guided recurrent learning, nonlinear temporal modeling, and uncertainty-aware recursive state estimation. Collectively, these components provide a robust and physically consistent framework for nonlinear state estimation in thermal power systems.

5.3. Leakage Verification Experiments

Given the high predictive accuracy achieved by the proposed framework, a series of verification experiments was performed to ensure that the reported performance does not arise from information leakage or inadvertent data contamination.

First, the complete dataset was partitioned chronologically into training (70%), validation (15%), and test (15%) subsets. All preprocessing operations, including normalization and sliding-window generation, were carried out only after the chronological split. The normalization statistics were computed exclusively from the training subset and then applied unchanged to the validation and test data. Moreover, the generated input sequences were constructed independently within each partition, preventing any temporal overlap between the training, validation, and test sets.

Second, a target-shuffling sanity check was conducted to verify that the proposed framework was learning genuine input-output relationships rather than exploiting hidden correlations. The training targets were randomly permuted, the linear N4SID state-space model was re-identified using only the shuffled training data, and the physics-informed loss was disabled $\lambda = 0$. Under these deliberately adversarial conditions, the resulting model produced negative coefficients of determination on the held-out test set, confirming that meaningful prediction cannot be achieved when the physical relationship between the inputs and outputs is destroyed.

Third, controlled ablation experiments were conducted to quantify the contribution of each component of the proposed framework. As shown in Tables 14 and 15, the Pure UKF baseline provides stable recursive state estimation but remains constrained by its linear process model, resulting in lower predictive accuracy than the proposed PI-LSTM-UKF across all monitored outputs. In particular, the coefficient of determination increases from 0.8323 to 0.9864 for the HP turbine valve, from 0.6951 to 0.9945 for the feedwater pump, from 0.8704 to 0.9854 for the steam extraction valve, from 0.8513 to 0.9967 for the system efficiency, and from 0.8432 to 0.9951 for the power output. In addition, replacing the recurrent Physics-Informed LSTM with a feedforward neural network (PI-NN-UKF) consistently reduces predictive performance. Although the PI-NN-UKF benefits from both physics-informed training and recursive Bayesian filtering, it fails to match the accuracy of the recurrent formulation, demonstrating that temporal memory is essential for modeling the long-term nonlinear dynamics of the thermal power system.

Collectively, these verification experiments support the conclusion that the reported performance is not attributable to information leakage or memorization. Instead, the observed improvements arise from the complementary integration of physics-informed recurrent learning and uncertainty-aware UKF-based recursive state estimation, implemented within a rigorously leakage-safe evaluation protocol.

5.4. Statistical Comparison Across Random Initializations

The effect of stochastic initialization was assessed by comparing the one-step-prior predictions of PI-LSTM-UKF and SINDy + LSTM over five matched random seeds. The PI-LSTM-UKF physics shares were fixed at $s_{\text{pre}} = 0.20$ and $s_{\text{fine}} = 0.50$. Within each pair, the two methods used the same simulated record, chronological data partitions, training-based preprocessing, 5% noise realization, and held-out test interval (samples 85,001–100,000).

The resulting comparisons therefore isolate variation due to neural network initialization and minibatch order under a fixed experimental setting. They should not be interpreted as tests across independent plant records or operating conditions.

The analysis was conducted directly from the full-precision results saved for each seed. To give both performance measures the same direction of interpretation, the R^2 difference was defined as

$$\Delta R_i^2 = R_{\mathrm{PI},i}^2 - R_{\mathrm{SINDy},i}^2, \tag{42}$$

and the normalized mean-squared-error difference as

$$\Delta \mathrm{MSE}_i = \mathrm{MSE}_{\mathrm{SINDy},i} - \mathrm{MSE}_{\mathrm{PI},i}. \tag{43}$$

A positive value consequently indicates better performance by PI-LSTM-UKF for either measure. Two-sided paired t-tests were applied to the five seed-wise differences, giving four degrees of freedom. Cohen's $d_z = \overline{\Delta}/s_\Delta$ was also reported to describe the standardized paired effect. To control the false discovery rate, the five output-specific p-values within each metric were adjusted using the Benjamini-Hochberg procedure. The across-output results were considered secondary summaries and were not included in this adjustment.

Before hypothesis testing, the numerical consistency of the paired data was checked for every output. In particular, we confirmed that

$$\overline{x_{\mathrm{PI}} - x_{\mathrm{SINDy}}} = \overline{x}_{\mathrm{PI}} - \overline{x}_{\mathrm{SINDy}}, \tag{44}$$

with the sign reversed under the MSE convention above. The equality held to double-precision accuracy, with an absolute discrepancy below 10^{-15} in all cases. The differences reported below were calculated from the original seed-level values. The displayed model means are rounded only for presentation.

As in Table 16 and 17, the comparison shows that neither method was uniformly better across all outputs. After adjustment for multiple testing, PI-LSTM-UKF achieved higher R^2 for the feedwater pump, system efficiency, and power, whereas SINDy + LSTM performed better for the HP turbine valve. The difference for the steam extraction valve was small and not statistically significant. These findings support an output-dependent interpretation rather than a general claim of superiority for either method.

Table 16. Paired comparison of noisy-reference one-step-prior R^2 for PI-LSTM-UKF ($s_{\mathrm{pre}} = 0.20$, $s_{\mathrm{fine}} = 0.50$) and SINDy + LSTM across five matched random seeds. Model values are mean ± sample standard deviation. The difference is $\Delta R^2 = R_{\mathrm{PI}}^2 - R_{\mathrm{SINDy}}^2$. Positive values favor PI-LSTM-UKF.

Output	PI-LSTM-UKF R^2	SINDy + LSTM R^2	Mean ΔR^2 (95% CI)
HP Turbine Valve	0.986153286 ± 0.000337	0.992624407 ± 0.000131	-0.0064711 $[-0.0068592, -0.0060830]$
Feedwater Pump	0.994520910 ± 0.000099	0.990582852 ± 0.000892	0.0039381 $[0.0027707, 0.0051055]$
Steam Extraction Valve	0.985065972 ± 0.000727	0.985289774 ± 0.001112	-0.0002238 $[-0.0020814, 0.0016337]$
System Efficiency	0.996651897 ± 0.000004	0.994234866 ± 0.000072	0.0024170 $[0.0023317, 0.0025023]$
Power (MW)	$0.995105069 \pm 0.000000014$	0.993713866 ± 0.000152	0.0013912 $[0.0012019, 0.0015805]$

The Benjamini-Hochberg adjustment was applied to the five output-specific R^2 tests.

As a secondary analysis, the five channel-wise values were averaged within each seed. The aggregate normalized-MSE improvement was 0.00064361 (95% CI $[0.00031805, 0.00096917]$, $t(4) = 5.489$, $p = 0.00537$, $d_z = 2.455$). In contrast, the mean channel-wise R^2 difference was 0.00021027 (95% CI $[-0.00038096, 0.00080150]$, $t(4) = 0.987$, $p = 0.379$, $d_z = 0.442$). This difference in aggregate significance is expected because an unweighted average of normalized MSE and an unweighted average of channel-wise R^2 do not weight the outputs in the same way. Each R^2 value is scaled by the variance of its

corresponding target. For this reason, the channel-specific results provide the clearest basis for interpreting relative performance.

Table 17. Paired comparison of noisy-reference one-step-prior normalized MSE across five matched random seeds. Model values are mean ± sample standard deviation. The improvement is $\Delta\text{MSE} = \text{MSE}_{\text{SINDy}} - \text{MSE}_{\text{PI}}$. Positive values favor PI-LSTM-UKF.

Output	PI-LSTM-UKF MSE	SINDy + LSTM MSE	Mean ΔMSE (95% CI)
HP Turbine Valve	$0.005022647 \pm 0.0001221$	$0.002675364 \pm 0.0000477$	-0.0023473 [$-0.0024881, -0.0022065$]
Feedwater Pump	$0.004383478 \pm 0.0000793$	$0.007534073 \pm 0.0007134$	0.0031506 [0.0022166, 0.0040846]
Steam Extraction Valve	$0.006349771 \pm 0.0003093$	$0.006254613 \pm 0.0004727$	-0.0000952 [-0.0008850, 0.0006947]
System Efficiency	$0.002504813 \pm 0.0000032$	$0.004313063 \pm 0.0000537$	0.0018083 [0.0017444, 0.0018721]
Power (MW)	$0.002468727 \pm 0.000000007$	$0.003170371 \pm 0.0000769$	0.0007016 [0.0006062, 0.0007971]

Output	$t(4)$	Raw p	BH p	d_z	Conclusion
HP Turbine Valve	-46.296	1.302×10^{-6}	3.255×10^{-6}	-20.704	SINDy + LSTM lower
Feedwater Pump	9.366	7.2385×10^{-4}	9.0481×10^{-4}	4.189	PI-LSTM-UKF lower
Steam Extraction Valve	-0.335	0.7548	0.7548	-0.150	No detected difference
System Efficiency	78.676	1.5643×10^{-7}	7.8215×10^{-7}	35.185	PI-LSTM-UKF lower
Power (MW)	20.403	3.4076×10^{-5}	5.6793×10^{-5}	9.125	PI-LSTM-UKF lower

The matching output-specific t statistics and p-values in the R^2 and MSE analyses follow from the relation $R^2 = 1 - \text{MSE}/c$ for a fixed output and test target, where c is the corresponding target-variance constant. Under the difference conventions used here, the two paired improvement vectors differ only by a positive scale factor.

5.5. Hyperparameter Sensitivity Analysis: Window Size and LSTM Hidden Units

To ensure reproducibility and robustness of the LSTM-based residual correction modules, a systematic sensitivity analysis was performed on two key hyperparameters: the sliding window size (`time_window`) and the number of hidden units in the LSTM layers. Four window sizes (10, 20, 30, 50) and three hidden unit configurations (64, 128, 256) were evaluated, resulting in 12 combinations.

Table 18 and Figure 14 present an illustrative sensitivity analysis of window size and the number of hidden units for the DMDc + LSTM residual corrector. This analysis was conducted under a separate representative operating scenario. The analysis reveals that:

- Window sizes of 10 provide insufficient temporal context, leading to notably higher MSE and lower R^2 values.
- Performance improves significantly for window sizes of 20 and above.
- The best performance is achieved with a window size of 50 and 256 hidden units ($\text{MSE}_{\text{Power}} = 0.00256$, $R^2_{\text{Power}} = 0.982$), while the combination of window size = 50 and hidden units = 128 offers an excellent trade-off between accuracy and computational efficiency.
- These configurations yield stable and high-accuracy results, especially for the critical outputs (power and feedwater pump).

The hyperparameters (window size = 50, hidden units = 128) lie in a stable high-performance region. Similar trends were observed for the SINDy + LSTM residual corrector and the PI-LSTM module in the closed-loop system (detailed results are provided in the Supplementary Materials).

Although the sensitivity analysis showed that a larger window size ($w = 50$) can improve the DMDc + LSTM residual corrector, the reported residual-learning results use $w = 20$ to reduce computational cost and maintain consistency with the CNN residual models. For the final PI-LSTM-UKF model, a window size of $w = 30$ is selected as a trade-off between temporal memory, computational cost, and the additional complexity introduced by UKF sigma-point propagation. Thus, $w = 50$ is reported only as part

of the hyperparameter sensitivity analysis, whereas $w = 20$ is used for the residual-learning benchmark results and $w = 30$ is used for the proposed closed-loop PI-LSTM-UKF framework.

Table 18. Sensitivity analysis of window size and LSTM hidden units for the DMDc + LSTM residual corrector. The best power R^2 value is shown in bold.

Window	Hidden	MSE					R^2				
		Valve	Feed	Steam	Eff	Power	Valve	Feed	Steam	Eff	Power
10	64	0.1062	0.00602	0.04300	0.01035	0.00985	0.8922	0.9396	0.9524	0.8936	0.9394
10	128	0.1024	0.00503	0.04224	0.00907	0.01026	0.8961	0.9496	0.9532	0.9068	0.9369
10	256	0.0747	0.00443	0.03647	0.00769	0.00838	0.9242	0.9555	0.9596	0.9209	0.9484
20	64	0.0703	0.00405	0.03095	0.00367	0.00432	0.9292	0.9594	0.9648	0.9616	0.9726
20	128	0.0715	0.00368	0.02696	0.00342	0.00373	0.9280	0.9631	0.9694	0.9642	0.9764
20	256	0.0900	0.00289	0.03732	0.00547	0.00666	0.9094	0.9709	0.9576	0.9429	0.9578
30	64	0.0437	0.00322	0.02754	0.00337	0.00431	0.9564	0.9675	0.9678	0.9641	0.9718
30	128	0.0562	0.00260	0.03377	0.00438	0.00408	0.9439	0.9738	0.9606	0.9533	0.9733
30	256	0.0576	0.00227	0.02974	0.00328	0.00432	0.9425	0.9771	0.9653	0.9651	0.9718
50	64	0.0454	0.00380	0.02740	0.00359	0.00415	0.9554	0.9615	0.9658	0.9602	0.9709
50	128	0.0326	0.00275	0.02598	0.00282	0.00314	0.9680	0.9721	0.9676	0.9688	0.9780
50	256	0.0398	0.00275	0.02242	0.00244	0.00256	0.9609	0.9721	0.9721	0.9730	**0.9820**

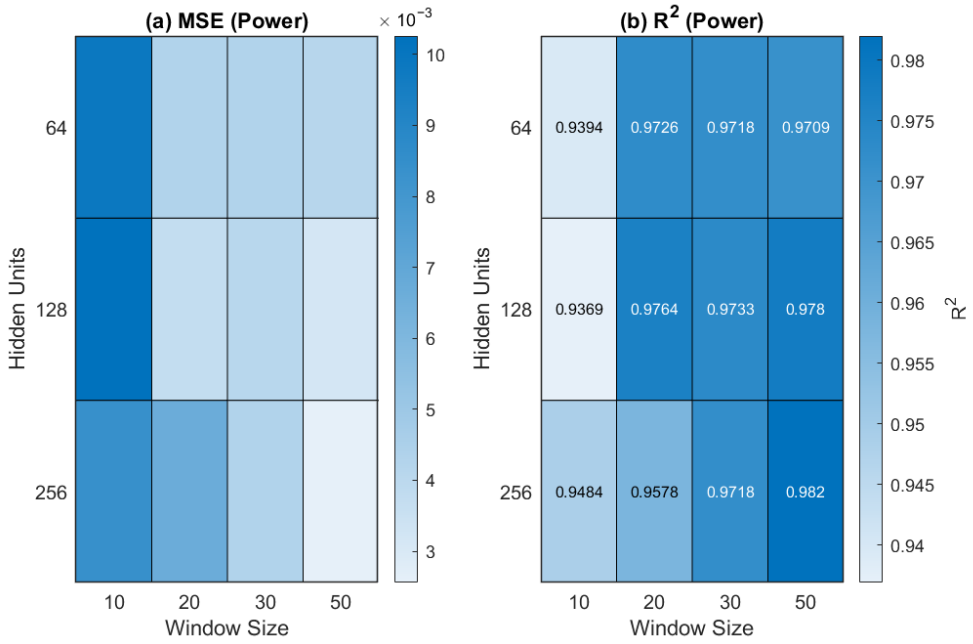


Figure 14. Heatmaps showing the effect of window size and number of LSTM hidden units on (**a**) MSE and (**b**) R^2 for the power (MW) output using the DMDc + LSTM residual corrector. Black and white numerical labels are used solely to maintain readability against the different heatmap background intensities.

The analysis was conducted in a separate representative operating scenario and independent training run and is intended only to assess hyperparameter trends. Therefore, its numerical values are not directly comparable with the five-run results reported in Tables 6 and 7.

5.6. Computational Efficiency and Inference Performance

To assess the computational efficiency of the proposed PI-LSTM-UKF framework, training overhead and online inference timing were measured on the hardware configuration summarized in Table 19. All experiments were conducted on an Intel Core i7 CPU with 32 GB RAM in the MATLAB/Simulink environment without GPU acceleration.

Table 19. Computational performance of the proposed PI-LSTM-UKF framework for evaluation of 100,000 samples.

Metric	Value
Hardware	Intel Core i7 CPU, 32 GB RAM
Number of evaluated samples	100,000
Total execution time	6566.14 s (1 h 49 min 26.14 s)
Non-inference time	3864.48 s (1 h 4 min 24.48 s)
Total UKF inference time	2701.66 s (45 min 1.66 s)
Average inference time	27.02 ms/sample
Inference throughput	37.01 samples/s

Table 19 reports the wall-clock cost of processing the full 100,000-sample record. The total execution time of 6566.14 s comprises a non-inference component of 3864.48 s (data handling, calibration, and related overhead) and a UKF inference component of 2701.66 s, which is equivalent to a throughput of 37.01 samples/s. For comparison, generating the high-fidelity Simscape reference trajectories required approximately 9–12 h.

Because the simulation sampling interval used throughout this study is 1 s, the measured latency of 27.02 ms/sample is substantially smaller than the sampling period.

The computational efficiency of the proposed PI-LSTM-UKF framework was evaluated in terms of model complexity, memory footprint, and inference speed. The number of trainable parameters and the corresponding model-weight memory were obtained directly from the trained network, while the computational complexity was estimated analytically based on the network architecture. For the adopted configuration (sequence length of 30, input dimension of 11, and stacked LSTM layers with 128 and 64 hidden units), a single forward pass requires approximately 3.63 million multiply-accumulate (MAC) operations, corresponding to approximately 7.27 million floating-point operations (FLOPs). Since the Unscented Kalman Filter propagates 11 sigma points for the selected five-dimensional latent state, the complete PI-LSTM-UKF prediction requires approximately 79.93 million FLOPs per prediction step, with the additional UKF covariance-update operations contributing only a negligible computational overhead. The runtime memory consumption and average inference time were measured experimentally after network warm-up to eliminate initialization overhead. These results indicate that the proposed framework achieves high prediction accuracy while maintaining a computational complexity compatible with near real-time digital twin deployment.

The framework is therefore computationally feasible for supervisory digital twin monitoring and condition assessment at the present 1 Hz update rate on standard CPU hardware. The current implementation was executed in the MATLAB interpreted environment without code generation or compiler optimization. Further reductions in latency are expected from MATLAB Coder, parallel execution, or GPU acceleration. For faster closed-loop control, deployment in compiled C/C++ or on embedded hardware would be required.

Evaluation of Physical Consistency

Conventional performance measures such as MSE and R^2 quantify agreement between predicted and simulated outputs, but do not directly establish compliance with the govern-

ing conservation laws. Physical consistency was therefore evaluated on the held-out test interval using residuals derived from the boiler and condenser mass and energy balances.

Because the compared models do not all expose the same physical state vector, a common decoder was identified using training data only. The decoder maps the five model outputs to the normalized physical state $\mathbf{s} = [M_b, E_b, M_c, E_c]^\mathsf{T}$, where M and E denote mass and energy and the subscripts b and c denote the boiler and condenser, respectively. The same decoder and normalization parameters were then applied without refitting to every model prediction on the held-out interval. For model m, the normalized one-step conservation residual was defined as

$$\mathbf{r}_k^{(m)} = \widehat{\mathbf{s}}_{k+1}^{(m)} - \mathcal{N}(\mathbf{s}_k + \Delta t\, \dot{\mathbf{s}}_k), \tag{45}$$

where $\widehat{\mathbf{s}}_{k+1}^{(m)}$ is the decoded state obtained from the prediction of model m, $\dot{\mathbf{s}}_k$ contains the mass- and energy-flow rates logged from the Simscape control volumes, and $\mathcal{N}$ denotes normalization using training-set statistics. The aggregate physics-residual RMSE (PRMSE) was calculated as

$$\mathrm{PRMSE}^{(m)} = \sqrt{\frac{1}{4N_{\mathrm{test}}} \sum_{k=1}^{N_{\mathrm{test}}} \left\| \mathbf{r}_k^{(m)} \right\|_2^2}. \tag{46}$$

Lower PRMSE values indicate smaller decoded violations of the conservation balances. Since this procedure uses an output-to-state decoder, the metric is interpreted as a common observable proxy for physical consistency rather than as a direct comparison of model-specific latent coordinates.

As shown in Table 20 and Figure 15 PI-LSTM-UKF achieved the lowest aggregate held-out conservation residual, with a PRMSE of 0.032862. The next-lowest value was 0.038527 for SINDy, hence, PI-LSTM-UKF reduced aggregate PRMSE by 14.7% relative to the strongest baseline. Its PRMSE reductions relative to DMDc, SINDy + CNN, DMDc + CNN, DMDc + LSTM, and SINDy + LSTM were 31.7%, 39.9%, 73.7%, 74.1%, and 29.4%, respectively.

PI-LSTM-UKF also produced the lowest residual in every normalized component. Relative to SINDy, which was the second-ranked model in aggregate PRMSE, the normalized boiler-mass, boiler-energy, condenser-mass, and condenser-energy residuals were reduced by 11.7%, 14.9%, 11.7%, and 9.9%, respectively. The physical-unit results in Table 21 preserve the same ordering: PI-LSTM-UKF attained residual RMSE values of 0.41250 kg, 2567.7 kJ, 0.41067 kg, and 97.367 kJ for boiler mass, boiler energy, condenser mass, and condenser energy, respectively.

Table 20. Held-out normalized conservation-residual comparison. Lower values indicate better decoded physical consistency. Bold indicates the best value in each column.

Model	PRMSE	Boiler Mass NRMSE	Boiler Energy NRMSE	Condenser Mass NRMSE	Condenser Energy NRMSE
DMDc	0.048147	0.015000	0.093405	0.015060	0.009802
SINDy	0.038527	0.012238	0.074636	0.012286	0.008130
DMDc + CNN	0.125090	0.044729	0.239940	0.044880	0.031724
SINDy + CNN	0.054673	0.020209	0.104520	0.020274	0.014595
DMDc + LSTM	0.126750	0.044516	0.243540	0.044670	0.031250
SINDy + LSTM	0.046570	0.015358	0.089962	0.015416	0.010408
PI-LSTM-UKF	**0.032862**	**0.010809**	**0.063494**	**0.010849**	**0.007328**

Table 21. Held-out mass- and energy-balance residuals in physical units. Bold indicates the best value in each column.

Model	Boiler Mass RMSE (kg)	Boiler Energy RMSE (kJ)	Condenser Mass RMSE (kg)	Condenser Energy RMSE (kJ)
DMDc	0.57244	3777.4	0.57007	130.24
SINDy	0.46706	3018.3	0.46507	108.03
DMDc + CNN	1.70700	9703.1	1.69880	421.53
SINDy + CNN	0.77126	4226.8	0.76743	193.93
DMDc + LSTM	1.69890	9848.9	1.69090	415.23
SINDy + LSTM	0.58613	3638.1	0.58353	138.30
PI-LSTM-UKF	**0.41250**	**2567.7**	**0.41067**	**97.367**

Interestingly, while residual CNN and LSTM corrections improved predictive accuracy relative to the pure DMDc and SINDy baselines, they increased the decoded conservation residuals. This indicates that purely data-driven residual learning can improve point-wise fit while partially relaxing the underlying mass- and energy-balance constraints. In contrast, the physics-informed PI-LSTM–UKF simultaneously improved both predictive accuracy and physical consistency.

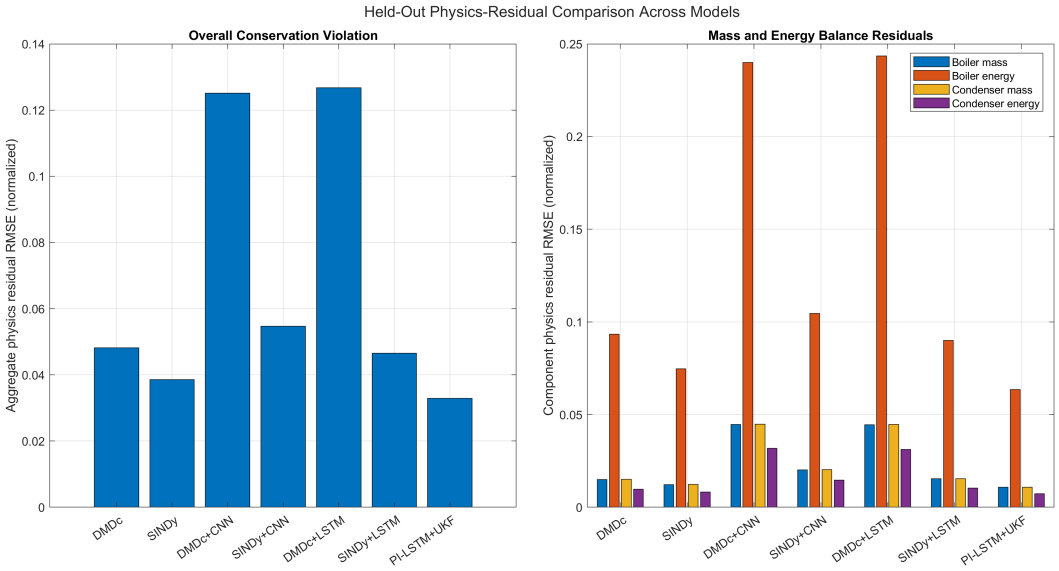


Figure 15. Held-out aggregate and component-wise normalized conservation residuals across the compared models. Lower bars indicate smaller decoded mass- and energy-balance violations.

These results provide held-out evidence that, under the common decoder, normalization, and evaluation protocol used here, the proposed PI-LSTM-UKF produces predictions that remain closer to the evaluated mass- and energy-conservation manifold than the six comparison models. This conclusion is restricted to the reported test interval and residual definition. Because the compared models do not expose identical physical states, the metric remains an observable, decoder-based proxy for conservation consistency and should not be interpreted as a direct equivalence between their internal state representations.

The proposed PI-LSTM model already contains a dedicated physical head that is trained to satisfy the conservation balances. The decoder-based residual metric defined above is introduced solely to enable a common comparison with models that do not expose internal physical states.

5.7. Comparison with Literature-Inspired Recurrent Baselines

To provide a controlled comparison with literature-inspired recurrent architectures, adapted BiLSTM-BiGRU-attention and physics-informed DeepLSTM (Phy-DeepLSTM) models were evaluated using the same experimental scenario as the main results. All

methods used the same continuous 100,000-sample Rankine-cycle record, 1 s sampling interval, Gaussian measurement-noise realization, chronological partitions, and held-out test interval. The record contained 70,000 training samples, 15,000 validation samples, and 15,000 test samples. Accordingly, the comparison used test indices 85,001–100,000.

For every method, normalization parameters were estimated exclusively from the training partition and applied unchanged to the validation and test partitions. The same 5%-noise realization, generated using data seed 42, was used as the evaluation reference. The table and prediction figures use the same target samples and test interval.

The adapted BiLSTM-BiGRU-attention model was inspired by the recurrent ensemble forecasting approach reported in [59] and served as a high-capacity data-driven recurrent baseline. The adapted Phy-DeepLSTM model was inspired by the physics-informed recurrent framework in [17]. Its physical regularization incorporated boiler and condenser mass- and energy-balance information.

For all methods, the reported quantity is a one-step-ahead prediction made without access to the contemporaneous target measurement y_k. The PI-LSTM-UKF result therefore corresponds to the prior prediction $\hat{y}_{k|k-1}$, evaluated before assimilation of y_k. The measurement-assisted UKF posterior was excluded because comparing it with predictors that do not observe y_k would not constitute an equivalent forecasting protocol.

Table 22 reports the cross-model R^2 comparison. To ensure a fair and consistent comparison, all methods (including the PI-LSTM-UKF prior) are reported using the same representative random seed. The curves shown in Figures 16 and 17 also correspond to the same seed.

Table 22. One-step-ahead normalized MSE and R^2 for the PI-LSTM-UKF prior and the literature-inspired recurrent baselines. All methods were evaluated against the same 5%-noise reference over test indices 85,001–100,000. Bold indicates the best value.

Output	PI-LSTM-UKF Prior		BiLSTM-BiGRU Attention		Phy-DeepLSTM	
	MSE	R^2	MSE	R^2	MSE	R^2
HP Turbine Valve	**0.004948**	**0.986358**	0.007471	0.979403	0.011216	0.969080
Feedwater Pump	0.004421	0.994474	0.009410	0.988238	**0.003560**	**0.995550**
Steam Extraction Valve	**0.006205**	**0.985407**	0.014330	0.966297	0.007025	0.983478
System Efficiency	**0.002505**	**0.996651**	0.007880	0.989467	0.026948	0.963979
Power (MW)	**0.002469**	**0.995116**	0.006342	0.987426	0.011161	0.977870
Mean across outputs	**0.004110**	**0.991599**	0.009087	0.982166	0.011982	0.977991

As shown in Table 22, PI-LSTM-UKF achieved the highest mean R^2 for four of the five outputs: the HP turbine valve, steam extraction valve, system efficiency, and power. For power, PI-LSTM-UKF achieved $R^2 = 0.995116$, compared with 0.987426 for BiLSTM-BiGRU attention and 0.977870 for Phy-DeepLSTM.

Phy-DeepLSTM achieved the highest feedwater pump R^2, with a value of 0.995550, compared with 0.994474 for PI-LSTM-UKF and 0.988238 for BiLSTM-BiGRU attention. The proposed method was therefore not superior on every output.

Across the five outputs, PI-LSTM-UKF achieved the highest mean R^2, equal to 0.991599. BiLSTM-BiGRU attention and Phy-DeepLSTM achieved mean R^2 values of 0.982166 and 0.977991, respectively. These results support the conclusion that PI-LSTM-UKF provided the strongest overall one-step predictive performance in this matched scenario.

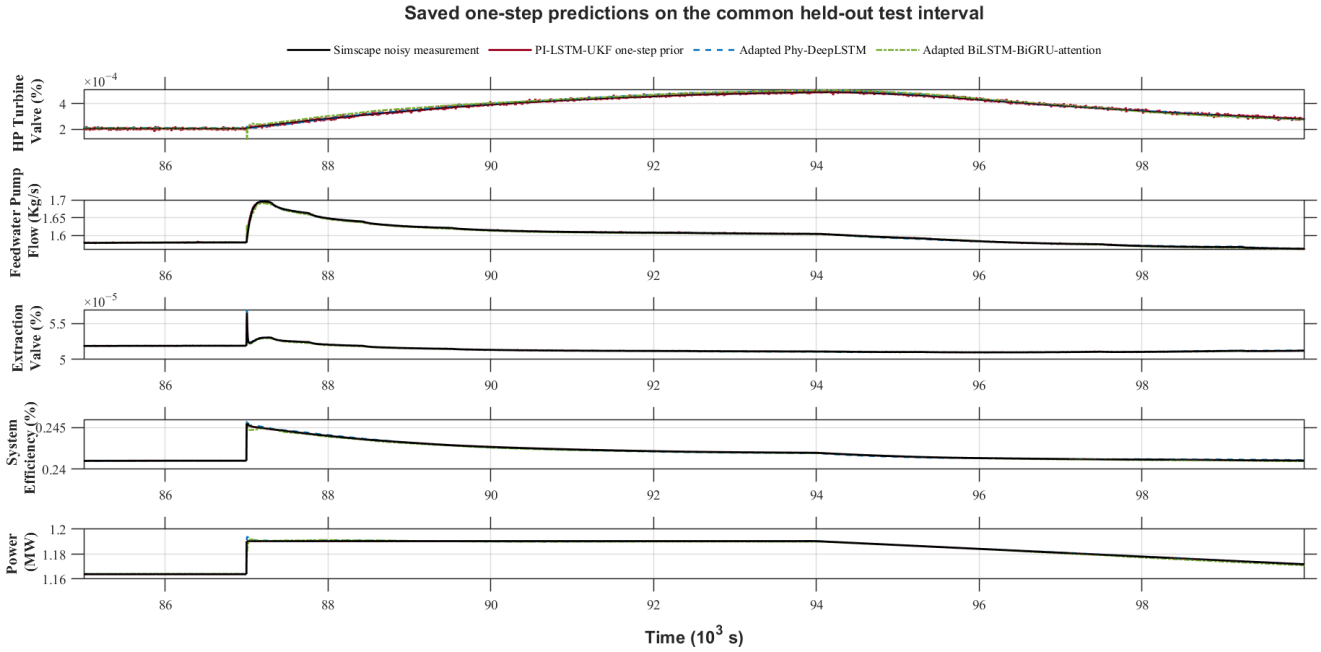


Figure 16. One-step-ahead predictions from the PI-LSTM-UKF prior and the two literature-inspired recurrent baselines on the same held-out test interval used in the main experiments. The black trace represents the common 5%-noise measurement reference. The PI-LSTM-UKF curve represents the prior before assimilation of the contemporaneous measurement.

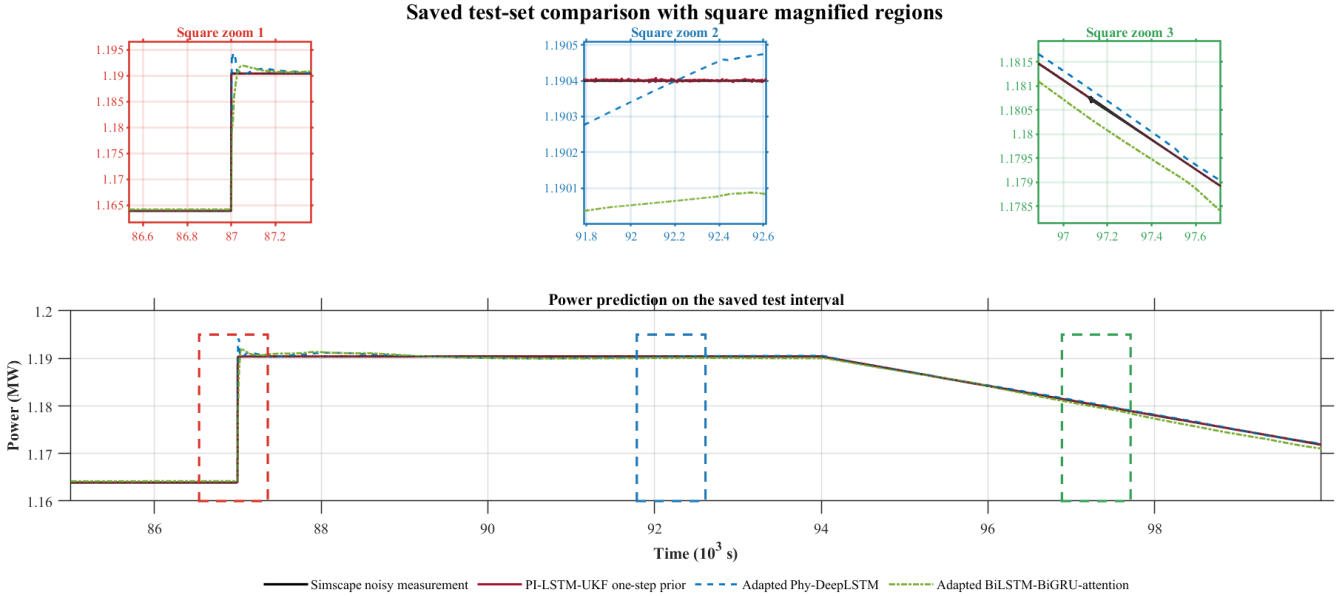


Figure 17. Magnified power regions from the common held-out test comparison. Consistent with Table 22, PI-LSTM-UKF achieved the highest overall power R^2.

5.8. Practical Implications and Comparison

The principal practical contribution of the proposed framework extends beyond improvements in prediction accuracy. Industrial digital twins must operate under noisy sensor measurements, changing operating conditions, and incomplete process knowledge while providing estimates that remain physically plausible. The combination of physics-informed recurrent learning and recursive Bayesian filtering addresses these practical requirements by integrating data-driven adaptability with thermodynamic consistency and uncertainty-aware state estimation.

From an operational perspective, the one-step prior provides a leakage-free prediction suitable for forecasting and anomaly detection, whereas the UKF posterior supplies measurement-assisted state estimates for real-time monitoring and supervisory decision support. Distinguishing these two operating modes is particularly important because the posterior incorporates the current measurement and therefore should be interpreted as recursive state estimation rather than independent forecasting.

The comparison with the literature-inspired recurrent baselines demonstrates that the proposed framework achieves competitive predictive accuracy while simultaneously providing uncertainty quantification and physically constrained learning. These additional capabilities are not reflected by R^2 alone but are important for practical digital twin deployment, where prediction reliability and physical consistency are often as important as pointwise accuracy as shown in Table 22.

Recent LSTM-based thermal-system models have reported R^2 values within the approximate range of 0.90–0.98 [60–62]. The proposed framework operates at the upper end of this range while additionally providing recursive uncertainty estimation and physics-guided predictions.

5.9. Knowledge-Guided Learning Interpretation

The proposed framework illustrates how simplified thermodynamic knowledge can guide deep recurrent learning without replacing the flexibility of data-driven modeling. Rather than enforcing a complete first-principles process model, the embedded thermodynamic constraints act as a regularizer that biases optimization toward physically plausible trajectories while allowing the recurrent network to capture nonlinear dynamics that are difficult to model analytically.

An additional advantage of this knowledge-guided formulation is that the learned latent dynamics become more interpretable than those of a purely black-box neural network. Although the latent states do not correspond directly to individual physical variables, the imposed conservation constraints encourage them to evolve in a manner that remains consistent with the underlying thermodynamic behavior of the plant. This improves the physical plausibility of the learned representation and reduces the likelihood of unrealistic predictions under off-design operating conditions.

The hierarchical comparison further demonstrates that linear modeling, residual learning, physics-informed regularization, and recursive Bayesian estimation each contribute different capabilities. Their integration therefore provides a balanced compromise between predictive accuracy, physical consistency, robustness to measurement noise, and uncertainty awareness, all of which are desirable characteristics for industrial digital twins.

5.10. Extremely Low Errors and Model Limitations

The high accuracy of the PI-LSTM-UKF should be interpreted in light of the simulation environment and preprocessing procedure. Outputs were normalized, reducing dynamic range, and generated from a high-fidelity Simscape model with smooth, thermodynamically consistent behavior. Although Gaussian noise was added, the simulated disturbances remain more bounded than those encountered in real plants, where sensor drift, fouling, aging, unmodeled hysteresis, and non-Gaussian disturbances introduce additional uncertainty. As noted in the ablation study, under degraded conditions, and noisy signals ($\sigma = 0.25$), performance of PI-LSTM-UKF decreased, indicating model sensitivity to thermodynamic mismatch. These results are therefore considered as an upper-bound benchmark and should not be interpreted as direct real-world performance expectations. Nevertheless, the inclusion of severe degradation scenarios, elevated sensor noise, hysteresis effects, and independently generated transient profiles provides a more challenging evaluation setting

than conventional simulation-only benchmark studies. Validation with real plant data is expected to reduce accuracy but strengthen robustness.

6. Conclusions

This work introduced a knowledge-guided physics-informed hybrid learning framework that integrates recurrent neural networks, embedded thermodynamic constraints, and Unscented Kalman Filter-based sequential state estimation for uncertainty-aware digital twin modeling of nonlinear thermal power systems. Using a two-stage training strategy, the framework enforces physical consistency within the learning objective while retaining the flexibility of data-driven models. A high-fidelity Rankine-cycle steam turbine model served as a challenging simulation-based case study.

Through systematic benchmarking against seven modeling variants (DMDc, SINDy, residual CNN and LSTM hybrids, and the full PI-LSTM-UKF), the study clarified the complementary contribution of each component. The UKF primarily improves estimation accuracy through recursive measurement correction, while the physics-informed loss regularizes the learned dynamics toward thermodynamic consistency. The resulting hybrid architecture achieved competitive one-step predictive accuracy while maintaining the lowest decoded conservation residuals among the compared models, demonstrating improved physical consistency without sacrificing predictive performance.

Ablation experiments further indicated that physics-informed training, recurrent modeling, and sequential filtering each make complementary contributions to predictive accuracy and physical consistency. Nevertheless, the paired comparison across random initializations showed that neither the proposed prior nor the strongest residual hybrid was uniformly superior across all outputs.

Although the experiments were conducted on high-fidelity simulated data, the framework demonstrated promising robustness under an independent unseen load profile with simultaneous component degradation and elevated measurement noise. These results should be interpreted as an upper-bound benchmark obtained under controlled conditions. Performance is expected to degrade on real plant data, where sensor drift, unmodeled hysteresis, and aging effects introduce additional challenges.

Future work will focus on validation with operational plant measurements, adaptive online learning, and integration into model predictive control and fault diagnosis frameworks. Overall, this study contributes to the growing body of research on physics-informed and hybrid machine learning methods for nonlinear industrial energy systems.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/make8080245/s1.

Author Contributions: Conceptualization, S.D. and A.S.A.-K.; methodology, S.D., M.M.E.-H. and A.S.A.-K.; software, S.D.; formal analysis, S.D.; investigation, S.D. and R.A.R.H.; data curation, S.D.; visualization, S.D.; validation, M.M.E.-H. and R.A.R.H.; resources, R.A.R.H.; writing—original draft preparation, S.D.; writing—review and editing, M.M.E.-H., A.S.A.-K. and R.A.R.H.; supervision, M.M.E.-H. and A.S.A.-K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Data will be available upon request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
PI	Physics Informed
DT	Digital Twin
ML	Machine Learning
N4SID	Numerical Algorithms for Subspace State-Space System Identification
SVD	Singular Value Decomposition
UKF	Unscented Kalman Filter
MAD	Median Absolute Deviation
DMDc	Dynamic Mode Decomposition with Control
SINDy	Sparse Identification of Nonlinear Dynamics
VARX	Vector AutoRegressive with eXogenous Inputs
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MSE	Mean Squared Error
R^2	Coefficient of Determination
L_{Boiler}	Boiler Drum Level
L_{heater}	Steam Pre-heat Level
$\dot{Q}_{\text{heat}}$	Heat Rate
$\dot{Q}_{\text{FW pump}}$	Feedwater Output Flow Rate
$V_{\text{super-heat}}$	Super-heat Valve Position
η_{system}	System Efficiency
$V_{\text{re-heat}}$	Re-heat Valve Position
V_{HP}	Actuator Position for High Pressure Valve
$V_{\text{extraction}}$	Actuator Position for Extraction Valve
P_{out}	Electrical Power Output
P_{demand}	Grid Power Demand

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